

The Creative Determinant: Autopoietic Closure as a Nonlinear Elliptic Boundary Value Problem with a Lean 4 Formalization

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Abstract

We present a formal mathematical realization of autopoietic closure in the sense of Maturana and Varela [1], expressed as a nonlinear elliptic boundary value problem on a compact Riemannian manifold. The *Creative Determinant* (CD) framework asks: *when can coherent presence sustain itself?* On a compact Riemannian manifold M (the *semiotic manifold*), we study the BVP with logistic saturation,

$$-\Delta\Phi = a(x) |\nabla\Phi| + b(x) \Phi - c(x) \Phi^p \quad \text{in } M, \quad \Phi = 0 \quad \text{on } \partial M,$$

where $a(x) = \kappa(x)\gamma(x)\mu(x) \in [0, 1]$ encodes the *creative drive* (care $\times$ coherence $\times$ contradiction), $b(x)$ is the *viability potential* (autopoietic support minus dissolution, with canonical closure $b = \kappa\gamma - \lambda\mu$), and $c(x)\Phi^p$ enforces *carrying capacity*. The principal eigenvalue condition $\lambda_1(-\Delta - b; M) < 0$ provides a computable criterion for the emergence of self-sustaining coherent configurations: it is sufficient in general and exact when the gradient coefficient vanishes—formalizing the transition from incoherence to autopoietic organization. We set up a Schaefer fixed-point framework with explicit a priori bounds for nonnegative *weak coherent configurations*, and prove by an ordered-barrier argument that when viability exceeds dissipation the BVP admits a solution that is positive throughout the interior, so coherent presence can be self-maintained. Beyond the PDE core, we distinguish weak from strong coherence (a gap we treat as the framework’s central generative tension), formulate conditional geometric statements, and propose an operational *Creative Determinant condition* $CD(\varepsilon, \delta)$ with five explicit falsifiability criteria. The core definitions, the algebraic and order-theoretic lemmas, and a finite-graph version of the positive-existence theorem with an explicit worked example are machine-checked in Lean 4 against the Mathlib library; the continuum existence theorems are formalized only as implications from an explicit abstract interface, and this paper gives an independent classical proof of the intended continuum results while recording that its concrete operator does not instantiate that interface. Numerical validation confirms second-order grid convergence and reproduces the verified finite-graph example. All proofs, computations, and verification artifacts are open source.

Keywords. autopoiesis; elliptic PDE; formal verification; Lean 4; fixed-point theorems; enactivism; dynamical systems; AI measurement.

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1 Introduction (interdisciplinary orientation)

Symbolic and subsymbolic approaches to cognition and computation are often treated as fundamentally distinct: symbols as discrete tokens vs. patterns as distributed fields. This paper develops a synthesis in which “symbolic” structure is represented as a continuous field over a manifold of meanings. The resulting framework—the *Creative Determinant* (CD)—formalizes the emergence and maintenance of coherent “presence” through the interaction of four ingredients: (i) gradient structure, (ii) care/adaptivity, (iii) coherence/consistency, and (iv) contradiction/curvature.

The manuscript is structured intentionally for an interdisciplinary audience while maintaining a rigorous mathematical spine:

- **Mathematical core (Sections 2–3).** A nonlinear elliptic PDE with logistic saturation on a compact Riemannian manifold, with a Schaefer/Leray–Schauder framework for nonnegative solutions, a positive-existence theorem using the eigenvalue threshold and ordered barriers, an exact threshold when the gradient term vanishes, and a machine-checked finite-graph analogue.
- **Interpretive/structural layer (Section 4).** A careful distinction between weak (PDE) and strong (pointwise) coherence; conditional geometric statements framed as explicit *ansätze* rather than theorems about the PDE.
- **Operational layer (Section 5).** A testable dynamical-systems condition and empirical falsification protocols, stated as proposals rather than proved consequences.

The central generative tension

A defining feature of this framework is the *necessary gap* between two notions of coherence:

- **Strong coherence:** the idealized pointwise constraint $\Phi = |\nabla\Phi|\kappa\gamma\mu$.
- **Weak coherence:** solutions to the associated PDE (1), which are mathematically tractable.

This gap is not a limitation—it is the *core philosophical commitment*. Real systems do not achieve idealized coherence; they *strive toward it* while settling into achievable configurations. The distance between strong and weak coherence represents the *work of living*—the ongoing effort required to approach but never fully reach ideal integration. This aligns with enactivist insights [2]: meaning is not a static state but an ongoing process of sense-making in the face of environmental perturbation.

Main mathematical contributions

- (i) **Existence framework (Theorem 3.12):** Under mild coefficient assumptions, the autonomous viability model with saturation has a compact fixed-point formulation with explicit a priori bounds for nonnegative weak coherent configurations; the zero field is always one of them.
- (ii) **Positive existence (Theorem 3.16):** When the viability threshold is exceeded ($\lambda_1(-\Delta - b; M) < 0$), there exists a solution positive throughout the interior, enclosed between a small multiple of the principal eigenfunction and a constant—coherent presence can be self-maintained.
- (iii) **Exact threshold without drive (Proposition 3.19):** When $a \equiv 0$ the spectral condition is also necessary, so the threshold is exact in that regime; with a gradient term it is sufficient only, and Proposition 3.21 gives a positive solution on an interval with $\lambda_1 = +\frac{1}{4}$, enclosed between an explicit subsolution and a constant supersolution.

- (iv) **Finite-graph theorem (Theorem 3.26):** A finite-graph version of the positive-existence theorem, proved without analytic hypotheses and machine-checked, with an explicit example, a maximum bound (Proposition 3.30), and a counterexample to the converse (Proposition 3.32).
- (v) **Canonical viability closure (Definition 3.3):** The viability potential $b = \kappa\gamma - \lambda\mu$ provides a principled link between the PDE coefficient and the CD characteristic fields.

Formal verification

The framework is accompanied by a Lean 4 development [3] against the Mathlib library [4, 5] (both at version v4.34.1), pinned to a specific revision in the companion repository. What is machine-checked: the definitions of the semiotic manifold model, coefficient structure, abstract operators, boundary value problem and weak coherent configuration; the coefficient bounds $0 \leq a \leq 1$; operator algebra; the algebraic core of the L^∞ bound (Lemma 3.10); a Knaster–Tarski fixed-point lemma, the order-theoretic skeleton of the sub-/super-solution method; the scaling argument excluding proportional solutions; the one-dimensional arithmetic of the threshold $\beta^* = (\pi/L)^2/b$; and, unconditionally, the finite-graph positive-existence theorem of Section 3.6 together with its explicit example. The continuum existence theorems (Theorems 3.12 and 3.16) are proved in Lean only as implications from the class `PDEInfra` and the structure `SolutionOperator`, whose fields stand in for classical elliptic results not available in Mathlib for manifolds. Those fields are hypotheses about an abstract operator, not Lean axioms, and they are not instantiated by the concrete operator of this paper; Appendix A records the mismatches. The analytic content of the continuum theorems is therefore the classical proof given in Section 3. Continuous integration builds every module with warnings treated as errors and audits the axioms of 49 selected declarations, all of which use only the standard Lean axioms. A complete alignment table is given in Appendix A; the numerical companion validates the finite-difference discretization with second-order grid convergence and reproduces the verified finite-graph example.

Main mathematical object and scope

The primary analytic object is the Dirichlet BVP with logistic saturation

$$\begin{cases} -\Delta\Phi = a(x) |\nabla\Phi| + b(x)\Phi - c(x)\Phi^p & \text{in } M, \\ \Phi = 0 & \text{on } \partial M, \end{cases} \quad (1)$$

where $a(x) := \kappa(x)\gamma(x)\mu(x) \in [0, 1]$, $b \in C^{0,\alpha}(M)$, $c(x) \geq c_0 > 0$, and $p > 1$. Solutions of (1) are called *weak coherent configurations*. The pointwise CD constraint $\Phi = |\nabla\Phi| \kappa\gamma\mu$ is treated as a constitutive idealization that motivates (1), but is not asserted equivalent; clarifying their relationship is an explicit open problem.

2 Mathematical setup

2.1 Semiotic manifold and function spaces

Definition 2.1 (Semiotic manifold). Let (M, g) be a compact, connected, smooth Riemannian manifold of dimension $n \geq 1$ with smooth boundary ∂M . For the Dirichlet problem (1) we require $\partial M \neq \emptyset$; the closed case $\partial M = \emptyset$ appears only in Proposition 4.11. We denote by ∇ the Riemannian gradient and by Δ the Laplace–Beltrami operator.

Remark 2.2 (Terminology: “semiotic”). The term *semiotic manifold* reflects the framework’s roots in Peircean semiotics [6], where meaning emerges through sign relations. The manifold M is interpreted as a space of possible meanings or interpretations.

We use standard Hölder spaces $C^{k,\alpha}(\overline{M})$ for $k \in \mathbb{N}$, $\alpha \in (0, 1)$, and Sobolev spaces $W^{k,q}(M)$, with the usual norms; see [7, 8]. The fixed-point arguments of Section 3 take place in

$$X := \{\psi \in C^1(\overline{M}) : \psi|_{\partial M} = 0\},$$

and we interpret $|\nabla\psi|$ pointwise using the Riemannian norm induced by g .

2.2 Core scalar fields and the coefficient a

Definition 2.3 (Characteristic fields). Let $\kappa, \gamma, \mu \in C^{0,\alpha}(M)$ be scalar fields with values in $[0, 1]$. Define

$$a(x) := \kappa(x)\gamma(x)\mu(x) \in [0, 1].$$

Remark 2.4 (Primary boundary condition). Dirichlet boundary conditions are taken as primary for analytic clarity and interpretive meaning: $\Phi = 0$ on ∂M represents boundary contexts where sustained presence/coherence is absent. Neumann conditions are discussed only as a secondary variant.

3 Main PDE Results

Guiding intuition. Before the formal machinery, here is the central question: *when can coherent presence sustain itself?* The answer involves a spectral condition—when the viability potential $b(x) = \kappa\gamma - \lambda\mu$ (care-coherence support minus contradiction cost) is strong enough to overcome boundary dissipation. This is made precise by the condition $\lambda_1(-\Delta - b; M) < 0$, which we call the *viability threshold*. The theorems below establish that crossing this threshold guarantees the existence of non-trivial coherent configurations. The proofs use classical fixed-point machinery, but the conceptual payoff is immediate: *viability exceeds dissipation $\Rightarrow$ coherent presence can be self-maintained*.

Figure 1 shows how the objects of this section fit together.

3.1 Primary model: autonomous viability with logistic saturation (V1’)

Let

$$a(x) := \kappa(x)\gamma(x)\mu(x) \in [0, 1]$$

be the CD modulation coefficient. We introduce a *viability potential* b and a *saturation (carrying capacity) field* c to obtain a mathematically nondegenerate autonomous equilibrium model.

Definition 3.1 (Autonomous viability model with saturation (V1’)). Fix $p > 1$. Given coefficients

$$a \in C^{0,\alpha}(M), \quad 0 \leq a \leq 1; \quad b \in C^{0,\alpha}(M); \quad c \in C^{0,\alpha}(M), \quad c(x) \geq c_0 > 0,$$

we consider the Dirichlet problem

$$\begin{cases} -\Delta\Phi = a(x)|\nabla\Phi| + b(x)\Phi - c(x)\Phi^p & \text{in } M, \\ \Phi = 0 & \text{on } \partial M. \end{cases} \quad (2)$$

A solution $\Phi \in C^{2,\alpha}(M)$ of (2) is called a *weak coherent configuration* for the autonomous CD model.

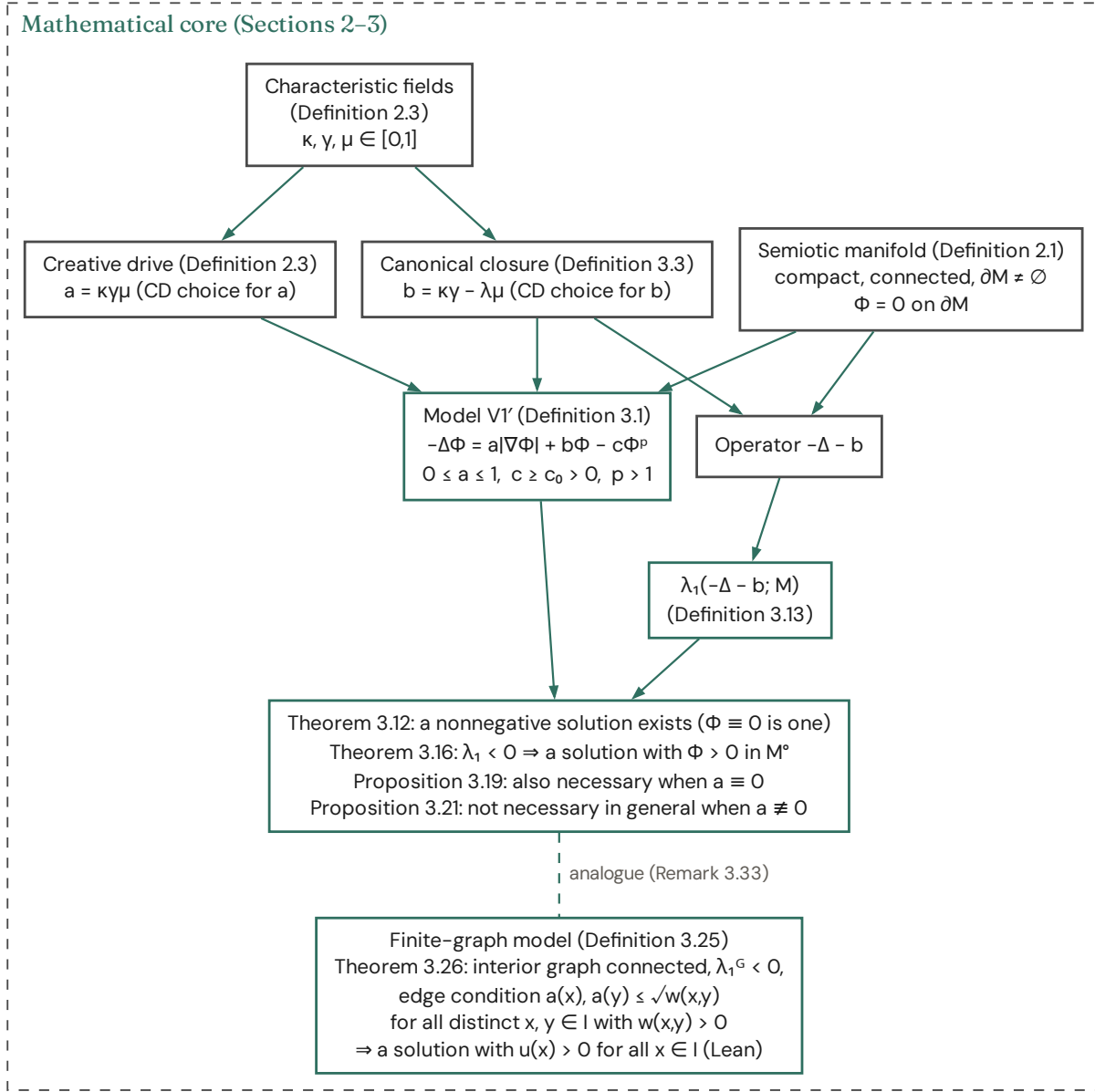


Figure 1: The mathematical core. The characteristic fields (Definition 2.3) define the creative drive $a = \kappa\gamma\mu$ and, through the canonical closure (Definition 3.3), the viability potential $b = \kappa\gamma - \lambda\mu$, which enter the saturated model V1' (Definition 3.1); the operator $-\Delta - b$ on M with Dirichlet data has the principal eigenvalue $\lambda_1(-\Delta - b; M)$ (Definition 3.13). Under the coefficient hypotheses of the model, a nonnegative weak coherent configuration exists and the zero field is one (Theorem 3.12); $\lambda_1 < 0$ gives a solution positive in the interior (Theorem 3.16); the condition is also necessary when $a \equiv 0$ (Proposition 3.19), and a counterexample shows that it is not necessary in general when $a \not\equiv 0$ (Proposition 3.21). The finite-graph model (Definition 3.25) is a separate problem (Remark 3.33); its existence theorem (Theorem 3.26), which assumes a connected interior graph, $\lambda_1^G < 0$ and the edge condition, is proved outright and machine-checked. Highlighted boxes are the model, the eigenvalue and the results.

Remark 3.2 (Interpretive glossary for (2)). In (2):

- $a(x) = \kappa\gamma\mu$ is the *creative drive* term: gradient activity contributes to presence only where care, coherence, and contradiction jointly support it.
- $b(x) > 0$ is *viability gain* / autopoietic support: presence amplifies where sustainable.
- $b(x) < 0$ is a *dissolution zone*: presence decays locally.
- $c(x)\Phi^p$ enforces *carrying capacity* / resource limitation: it prevents unbounded growth and encodes finite-resource autopoiesis.

Definition 3.3 (Canonical viability closure). Under the *canonical viability closure*, the viability potential is defined by

$$b(x) := \kappa(x)\gamma(x) - \lambda\mu(x), \quad (3)$$

where $\lambda > 0$ is a contradiction-weighting parameter.

Remark 3.4 (Empirical motivation for the linear closure). The canonical closure $b = \kappa\gamma - \lambda\mu$ is not derived from first principles, but it is not arbitrary either. It represents the *simplest mass-action balance* reflecting a fundamental trade-off observed across biological, psychological, and social systems: **support** (care $\times$ coherence) competes against **contradiction cost** ($\lambda \times$ contradiction level).

This structure echoes the “recovery capital” framework in behavioral health [9, 10]: sustained wellbeing emerges when supportive resources outweigh risk factors. The linear form is the minimal assumption—the V1 approximation—before introducing higher-order effects. Alternative closures (multiplicative, exponential, threshold-based) remain open for investigation, but the linear closure captures the essential bifurcation structure: when support dominates cost, nontrivial configurations exist; when cost dominates, the system collapses to triviality.

Seven years of observation in peer support and recovery research informed this choice: systems that sustain coherence reliably exhibit this support-versus-cost dynamic. The math formalizes what experience revealed.

Remark 3.5 (The contradiction-cost parameter λ). The parameter λ is not merely a tuning knob—it is an *empirically observable characteristic* of a system’s sensitivity to internal contradiction. Crucially, in the closure (3), λ functions as a **cost multiplier**: increasing λ decreases b for fixed $\mu > 0$, making the spectral condition $\lambda_1(-\Delta - b; M) < 0$ harder to satisfy.

- A system with **high** λ (contradiction-sensitive) experiences contradiction as costly: even moderate μ significantly reduces viability. Example: a brittle belief system that fragments when faced with counter-evidence.
- A system with **low** λ (contradiction-resilient) can sustain coherence even when μ is elevated, because contradiction exerts less drag on viability. Example: a healthy cognitive system that integrates conflicting information without destabilizing.

This interpretation provides a concrete empirical program: estimate λ from observed bifurcation points where systems transition from coherent to incoherent behavior. Systems at the “cliff edge” of viability (small negative λ_1) are those where care-coherence support barely exceeds contradiction cost—a measurable condition of precarious stability.

We emphasize that the closure parameter λ in (3) is distinct from the principal eigenvalue $\lambda_1(-\Delta - b; M)$: the former weights contradiction in the viability potential, while the latter is a spectral indicator of whether nontrivial coherence can exist (Theorem 3.16).

Remark 3.6 (Limiting case: the homogeneous model (V1)). If $c \equiv 0$, (2) reduces to the positively homogeneous equation

$$-\Delta\Phi = a(x)|\nabla\Phi| + b(x)\Phi, \quad \Phi|_{\partial M} = 0,$$

which is an idealized “no-saturation” limit. The saturation term is mathematically and interpretively important: it breaks homogeneity and enables barrier constructions yielding nontrivial equilibria.

3.2 Existence of nonnegative weak coherent configurations (Schaefer/Leray–Schauder)

We now set up the fixed-point framework for (2). Throughout, $\partial M \neq \emptyset$ (the Dirichlet problem needs a boundary), $n = \dim M$, and $q > n$ is a fixed exponent. The results cited from Gilbarg–Trudinger [7] are stated there for bounded domains in $\mathbb{R}^n$ whose boundary is $C^{1,1}$ (Theorems 9.15 and 7.28, Lemma 9.17), $C^{2,\alpha}$ (Theorem 6.14), Lipschitz (Theorem 7.26) or unrestricted (Theorem 3.1, Corollary 3.2, Theorem 3.5); a smooth boundary satisfies each of these. They are not applied to M directly. Appendix B records how each is transferred to the compact manifold: in every chart the Laplace–Beltrami operator is uniformly elliptic with smooth coefficients, so the Sobolev embedding, the interpolation inequality and the interior and boundary $W^{2,q}$ and Schauder *estimates* transfer by a finite partition of unity; *existence* for the linear Dirichlet problems $-\Delta u + Ku = f$, $K \geq 0$, is obtained from the Lax–Milgram theorem in $H_0^1(M)$ followed by H^2 , $W^{2,q}$ and $C^{2,\beta}$ regularity in charts (and, for $1 < q < 2$, which can occur only when $n = 1$, by approximation of the datum in L^q together with the a priori estimate), with uniqueness from Green’s formula; and the *weak maximum principle* on an open subset of M° is derived from the strong maximum principle [7, Thm. 3.5], which is local, together with the connectedness of M .

What changed. An earlier version of this section worked with the operator $\tilde{T}(\psi) = S(a|\nabla\psi| + b\psi - c(\psi_+)^p)$ in Hölder spaces and asserted a uniform bound $\|u\|_{L^\infty} \leq K$ for its Schaefer set together with nonnegativity of its fixed points. Both assertions are false: on $M = (0, \pi)$ with $a = 0$, $b = c = 1$, $p = 2$, every $u_A(x) = -A \sin x$, $A > 0$, satisfies $-u_A'' = u_A = u_A - (u_A)_+^2$, hence $\tilde{T}(u_A) = u_A$, so $\tilde{T}$ has negative fixed points of arbitrarily large norm. The repair below truncates the linear term as well, works in C^1 rather than $C^{1,\alpha}$ (the map $\psi \mapsto |\nabla\psi|$ is bounded but not continuous in Hölder norm), and obtains nonnegativity from the weak maximum principle before the L^∞ bound.

Operator formulation

Let

$$X := \{\psi \in C^1(\overline{M}) : \psi|_{\partial M} = 0\}, \quad \|\psi\|_X := \|\psi\|_{C^1(\overline{M})},$$

a Banach space. For $\psi \in X$ define the *truncated nonlinearity*

$$F(\psi)(x) := a(x)|\nabla\psi(x)| + b(x)\psi_+(x) - c(x)(\psi_+(x))^p, \quad \psi_+ := \max\{\psi, 0\}.$$

For $\psi \geq 0$ this is the right-hand side of (2). Let S denote the Dirichlet solution operator of $-\Delta$: for $f \in L^q(M)$ there is a unique $u = Sf \in W^{2,q}(M) \cap W_0^{1,q}(M)$ with $-\Delta u = f$, and $\|Sf\|_{W^{2,q}} \leq C\|f\|_{L^q}$ [7, Thm. 9.15, Lem. 9.17]. Define

$$T : X \rightarrow X, \quad T(\psi) := S(F(\psi)).$$

Lemma 3.7 (Compactness). *$T : X \rightarrow X$ is well defined, continuous, and compact.*

Proof. The maps $\xi \mapsto |\xi|$ on $\mathbb{R}^n$, $s \mapsto s_+$ and $s \mapsto (s_+)^p$ on $\mathbb{R}$ are continuous, so $\psi \mapsto F(\psi)$ is continuous from X to $C^0(\overline{M})$ (uniformly on bounded sets) and bounded: $\|F(\psi)\|_\infty \leq \|\nabla\psi\|_\infty + \|b\|_\infty\|\psi\|_\infty + \|c\|_\infty\|\psi\|_\infty^p$. Composing with the bounded operator $S : L^q \rightarrow W^{2,q}$ and the Sobolev embedding $W^{2,q}(M) \hookrightarrow C^{1,\gamma}(\overline{M})$, $\gamma = 1 - n/q$ [7, Thm. 7.26], shows that T is continuous into $C^{1,\gamma}(\overline{M})$ and maps bounded sets of X to bounded sets of $C^{1,\gamma}(\overline{M})$. Bounded sets of $C^{1,\gamma}(\overline{M})$ are relatively compact in $C^1(\overline{M})$ (Arzelà–Ascoli), and $T(\psi)$ vanishes on ∂M . Hence $T : X \rightarrow X$ is continuous and compact. $\square$

A priori bounds for Schaefer

Schaefer’s theorem ([8, §9.2.2, Thm. 4]; [11, 12]) gives a fixed point of a continuous compact self-map of a Banach space once the set

$$\mathcal{S} := \{u \in X : u = \tau T(u) \text{ for some } \tau \in [0, 1]\}$$

is bounded. Elements of $\mathcal{S}$ satisfy $u \in W^{2,q}(M)$ and

$$-\Delta u = \tau F(u) = \tau \left(a|\nabla u| + b u_+ - c(u_+)^p \right) \text{ a.e. in } M, \quad u|_{\partial M} = 0. \quad (4)$$

Assumption 3.8 (Upper bound on b). Let $B \geq 0$ satisfy $b(x) \leq B$ for all $x \in M$. Since $b \in C^{0,\alpha}(M)$ and M is compact, $B := \max\{\max_M b, 0\}$ always works; the assumption only names the constant that enters the bounds below.

Lemma 3.9 (Regularity and sign of the Schaefer set). *Every $u \in \mathcal{S}$ belongs to $C^{2,\gamma'}(\overline{M})$ with $\gamma' := \min\{\alpha, \gamma\}$ and satisfies $u \geq 0$ on M .*

Proof. Let $u = \tau T(u)$. By the Sobolev embedding $u \in C^{1,\gamma}(\overline{M})$, so $|\nabla u|$, u_+ and $(u_+)^p$ are in $C^{0,\gamma}(\overline{M})$ (compositions with the Lipschitz maps $|\cdot|$, $(\cdot)_+$, and the locally Lipschitz map $s \mapsto (s_+)^p$, $p > 1$). Thus $\tau F(u) \in C^{0,\gamma'}(\overline{M})$, and the Dirichlet problem $-\Delta v = \tau F(u)$, $v|_{\partial M} = 0$, has a solution $v \in C^{2,\gamma'}(\overline{M})$ [7, Thm. 6.14]; by uniqueness in $W^{2,q} \cap W_0^{1,q}$, $v = u$.

For the sign, let $U := \{x \in M^\circ : u(x) < 0\}$, an open set, and suppose $U \neq \emptyset$. On U , $u_+ = 0$, so $-\Delta u = \tau a|\nabla u| \geq 0$ pointwise. Put $\beta(x) := \tau a(x)\nabla u(x)/|\nabla u(x)|$ where $\nabla u(x) \neq 0$ and $\beta(x) := 0$ otherwise; β is bounded and measurable with $|\beta| \leq 1$, and $\tau a|\nabla u| = \beta \cdot \nabla u$. Then $w := -u \in C^2(U) \cap C^0(\overline{U})$ satisfies $\Delta w + \beta \cdot \nabla w = 0$ in U , $w > 0$ in U , and $w = 0$ on ∂U (on ∂M because $u|_{\partial M} = 0$, and elsewhere on ∂U by continuity). The weak maximum principle for elliptic operators with bounded measurable coefficients [7, Thm. 3.1], in the form valid on open subsets of M° (Appendix B(e), via [7, Thm. 3.5]), gives $\sup_U w = \sup_{\partial U} w = 0$, a contradiction. Hence $U = \emptyset$ and $u \geq 0$. $\square$

Lemma 3.10 (Uniform L^∞ bound). *Under Assumption 3.8, every $u \in \mathcal{S}$ satisfies*

$$0 \leq u \leq K := \left(\frac{B}{c_0} \right)^{1/(p-1)} \text{ on } M.$$

More generally, every nonnegative solution $\Phi \in C^2(\overline{M})$ of (2) satisfies $\max_M \Phi \leq K$.

Proof. By Lemma 3.9, $u \geq 0$ and u is a classical solution of (4), in which now $u_+ = u$. If $\max_M u = 0$ there is nothing to prove, and if $\tau = 0$ then $u = 0$. Otherwise the maximum is attained at $x_0 \in M^\circ$ (since $u = 0$ on ∂M), where $\nabla u(x_0) = 0$ and $-\Delta u(x_0) \geq 0$. Evaluating (4) at x_0 ,

$$0 \leq \tau(b(x_0)u(x_0) - c(x_0)u(x_0)^p) \leq \tau(Bu(x_0) - c_0u(x_0)^p),$$

and since $\tau > 0$ and $u(x_0) > 0$, this gives $u(x_0)^{p-1} \leq B/c_0$. The same evaluation with $\tau = 1$ proves the statement for nonnegative classical solutions. (The algebraic step from $Bv \geq c_0 v^p$ to $v \leq (B/c_0)^{1/(p-1)}$ is the Lean lemma `linfty_bound_algebraic`; see Appendix A.) $\square$

Lemma 3.11 (Uniform C^1 bound). *Under Assumption 3.8, $\mathcal{S}$ is bounded in X ; in fact it is bounded in $C^{1,\gamma}(\overline{M})$.*

Proof. Let $u \in \mathcal{S}$. By Lemma 3.10, $0 \leq u \leq K$, so $|bu| \leq \|b\|_\infty K$ pointwise (the upper bound B controls the maximum but not $|b|$, which may be large where b is negative), and $|\tau F(u)| \leq |\nabla u| + C_0$ with $C_0 := \|b\|_\infty K + \|c\|_\infty K^p$. The $W^{2,q}$ estimate for the Dirichlet problem [7, Lem. 9.17] and the interpolation inequality [7, Thm. 7.28] give, for every $\varepsilon > 0$,

$$\|u\|_{W^{2,q}} \leq C\|\tau F(u)\|_{L^q} \leq C(\|\nabla u\|_{L^q} + C_0|M|^{1/q}) \leq C(\varepsilon\|u\|_{W^{2,q}} + C_\varepsilon\|u\|_{L^q} + C_0|M|^{1/q}).$$

Choosing $\varepsilon = 1/(2C)$ and using $\|u\|_{L^q} \leq K|M|^{1/q}$ yields $\|u\|_{W^{2,q}} \leq C'$, with C' depending only on $M, q, B, c_0, p, \|b\|_\infty$ and $\|c\|_\infty$. The Sobolev embedding then bounds $\|u\|_{C^{1,\gamma}}$, hence $\|u\|_X$. $\square$

Theorem 3.12 (Existence of a nonnegative weak coherent configuration). *Assume the coefficient hypotheses of Definition 3.1. Then T has a fixed point $\Phi \in X$; every fixed point of T is a nonnegative solution $\Phi \in C^{2,\alpha}(\overline{M})$ of (2), and every nonnegative classical solution satisfies $\max_M \Phi \leq (B/c_0)^{1/(p-1)}$.*

Proof. By Lemmas 3.7 and 3.11, Schaefer's theorem yields $\Phi \in X$ with $T(\Phi) = \Phi$, i.e. $\Phi \in \mathcal{S}$ with $\tau = 1$. By Lemma 3.9, $\Phi \geq 0$ and $\Phi \in C^{2,\gamma'}(\overline{M})$; since $\Phi_+ = \Phi$, Φ solves (2). One more Schauder step upgrades the regularity: $\nabla \Phi \in C^{1,\gamma'}$ makes $|\nabla \Phi|$ Lipschitz, so the right-hand side of (2) lies in $C^{0,\alpha}(\overline{M})$ and $\Phi \in C^{2,\alpha}(\overline{M})$ [7, Thm. 6.14]. The bound is Lemma 3.10. $\square$

The zero solution. $\Phi \equiv 0$ solves (2) for every admissible choice of coefficients (Lean: `zero_solves_equation`), so Theorem 3.12 does not by itself produce coherent presence. Its content is the fixed-point framework and the a priori bounds, which Theorem 3.16 reuses; the substantive question is the existence of a solution that is positive in the interior.

3.3 Viability exceeds dissipation: the condition for non-trivial coherence

The existence theorem above guarantees a nonnegative solution, and the zero function is one. The following theorem provides the key criterion: when does the system support *genuine* coherent presence? The answer is a spectral condition on the viability potential—precisely, when integrated care-coherence support dominates contradiction load strongly enough to overcome boundary dissipation.

Definition 3.13 (Principal eigenvalue and viability threshold). Let $\lambda_1(-\Delta - b; M)$ denote the principal Dirichlet eigenvalue of the operator $\mathcal{L}_b u := -\Delta u - b(x)u$ on M . We say the *viability threshold is exceeded* if

$$\lambda_1(-\Delta - b; M) < 0.$$

Remark 3.14 (Spectral theory). For the symmetric operator $\mathcal{L}_b$ with Dirichlet conditions, λ_1 is simple and has an eigenfunction ϕ_1 that is positive in M° (M is connected): this is the statement for self-adjoint divergence-form operators with bounded measurable coefficients [7, Thm. 8.38], transferred to M as in Appendix B(f); the variational proof is exposed for smooth coefficients and no zero-order term in [8, §6.5.1, Thm. 2] and extends to the bounded potential $-b$ by the same

argument once $C^{2,\alpha}$ regularity of the minimizer is available (Appendix B(d)). For nonsymmetric operators the corresponding statement is [8, §6.5.2, Thm. 3] and the Krein–Rutman theorem [13]. Since $b \in C^{0,\alpha}(M)$, Schauder regularity gives $\phi_1 \in C^{2,\alpha}(\overline{M})$.

Remark 3.15 (Interpretation of the viability threshold). Under the canonical closure $b = \kappa\gamma - \lambda\mu$ (Definition 3.3), the spectral condition $\lambda_1(-\Delta - b; M) < 0$ becomes a precise “*viability exceeds dissipation*” criterion: integrated care–coherence support dominates contradiction load strongly enough to overcome boundary dissipation imposed by $\Phi|_{\partial M} = 0$.

Theorem 3.16 (Existence of a positive weak coherent configuration). *Assume the coefficient hypotheses of Definition 3.1 and suppose*

$$\lambda_1(-\Delta - b; M) < 0.$$

Let ϕ_1 be the principal eigenfunction normalized by $\|\phi_1\|_{L^\infty} = 1$, and set

$$\varepsilon := \min \left\{ 1, \left(\frac{-\lambda_1}{\|c\|_\infty} \right)^{1/(p-1)} \right\}, \quad M_* := \max \left\{ 1, \left(\frac{B}{c_0} \right)^{1/(p-1)} \right\}.$$

Then (2) admits a solution $\Phi \in C^{2,\alpha}(\overline{M})$ with

$$\varepsilon \phi_1 \leq \Phi \leq M_* \quad \text{in } M; \quad \text{in particular } \Phi > 0 \text{ in } M^\circ.$$

Proof. Write $g(x, s) := b(x)s - c(x)s^p$ for $s \geq 0$ and $\beta := -\lambda_1 > 0$, so that $-\Delta\phi_1 = b\phi_1 - \beta\phi_1$.

Step 1: ordered barriers. *Subsolution.* Let $v := \varepsilon\phi_1$. Then $v = 0$ on ∂M , $v > 0$ in M° , and

$$-\Delta v = \varepsilon b\phi_1 - \varepsilon\beta\phi_1.$$

A subsolution has to satisfy $v \leq 0$ on ∂M (here $v = 0$, the prescribed data) and the differential inequality $-\Delta v \leq a|\nabla v| + g(x, v)$ in M° ; only these two properties are used in Step 3. (On ∂M the left side vanishes while the right side equals $\varepsilon a|\nabla\phi_1| \geq 0$, which is generally nonzero, so the inequality extends to the boundary but not as an equality.) In M° the inequality reads, after cancelling $\varepsilon b\phi_1$ and dividing by $\varepsilon\phi_1 > 0$,

$$-\beta \leq a \frac{|\nabla\phi_1|}{\phi_1} - c\varepsilon^{p-1}\phi_1^{p-1}.$$

Since $a|\nabla\phi_1| \geq 0$ and $0 < \phi_1 \leq 1$, it suffices that $\|c\|_\infty \varepsilon^{p-1} \leq \beta$, which is the choice of ε .

Supersolution. The constant $w := M_*$ satisfies $w \geq 0 = \Phi|_{\partial M}$ on ∂M and $-\Delta w = 0$, $|\nabla w| = 0$, so $-\Delta w \geq a|\nabla w| + g(x, w)$ is equivalent to $c(x)M_*^{p-1} \geq b(x)$, which follows from $c_0 M_*^{p-1} \geq B$. A supersolution need only dominate the zero boundary data; it need not vanish on ∂M .

Ordering. $v \leq \varepsilon \leq 1 \leq M_* = w$.

What Steps 2–4 use. Only the following properties of the pair (v, w) enter below: $v \in C^2(\overline{M})$, $v \geq 0$, $v = 0$ on ∂M , the subsolution inequality in M° ; w a constant with $w \geq \max v$ and $g(x, w) \leq 0$ for all x . Any pair with these properties can replace $(\varepsilon\phi_1, M_*)$, which is how Proposition 3.21 below is proved.

Step 2: a truncated problem. Let $\theta(x, s) := \min\{\max\{s, v(x)\}, M_*\} \in [v(x), M_*]$ and choose

$$K := 1 + \max\left\{0, \|c\|_\infty p M_*^{p-1} - \min_M b\right\},$$

so that $s \mapsto g(x, s) + Ks$ is nondecreasing on $[0, M_*]$ for every x (its derivative is $b - c p s^{p-1} + K \geq 1$). Consider

$$-\Delta u + Ku = a|\nabla u| + h(x, u) \text{ in } M, \quad u|_{\partial M} = 0, \quad h(x, s) := g(x, \theta(x, s)) + K\theta(x, s). \quad (5)$$

The function h is continuous and bounded, $|h| \leq H := BM_* + \|c\|_\infty M_*^p + KM_*$. Let S_K be the Dirichlet solution operator of $-\Delta + K$, bounded $L^q \rightarrow W^{2,q} \cap W_0^{1,q}$ with $\|S_K f\|_{W^{2,q}} \leq C\|f\|_{L^q}$ [7, Thm. 9.15, Lem. 9.17] (the zero-order coefficient $-K$ is nonpositive), and define $T_K(\psi) := S_K(a|\nabla \psi| + h(\cdot, \psi))$ on X . Exactly as in Lemma 3.7, $T_K : X \rightarrow X$ is continuous and compact.

Let $u = \tau T_K(u)$, $\tau \in [0, 1]$. As in Lemma 3.9, $u \in C^{1,\gamma}(\overline{M})$; moreover $\theta(\cdot, u(\cdot))$ is a minimum and maximum of Hölder functions, hence Hölder, and g is C^1 in s with Hölder coefficients, so $h(\cdot, u(\cdot)) \in C^{0,\gamma'}(\overline{M})$ and $u \in C^{2,\gamma'}(\overline{M})$ is a classical solution of $-\Delta u + Ku = \tau(a|\nabla u| + h(x, u))$. At a positive interior maximum x_0 of u , $\nabla u(x_0) = 0$ and $-\Delta u(x_0) \geq 0$, whence $Ku(x_0) \leq \tau H \leq H$; at a negative interior minimum, $Ku \geq -H$. Hence $\|u\|_\infty \leq H/K$, and the argument of Lemma 3.11 (with $|\tau(a|\nabla u| + h)| \leq |\nabla u| + H$) bounds the Schaefer set of T_K in X . Schaefer's theorem gives a fixed point $u \in X$ of T_K , which is a classical solution of (5) in $C^{2,\gamma'}(\overline{M})$.

Step 3: comparison with the barriers. Set $U := \{x \in M^\circ : u(x) < v(x)\}$, open, and suppose $U \neq \emptyset$. On U we have $u < v \leq M_*$, hence $\theta(x, u) = v(x)$ and $h(x, u) = g(x, v) + Kv$. Using the subsolution inequality for v , the difference $z := v - u \in C^2(U) \cap C^0(\overline{U})$ satisfies in U

$$-\Delta z + Kz \leq a|\nabla v| + g(x, v) + Kv - (a|\nabla u| + g(x, v) + Kv) = a(|\nabla v| - |\nabla u|) \leq a|\nabla z|.$$

With $\beta := a|\nabla z|/|\nabla z|$ where $\nabla z \neq 0$ (and 0 elsewhere), this is $\Delta z + \beta \cdot \nabla z - Kz \geq 0$ in U , an inequality for a uniformly elliptic operator with bounded coefficients and nonpositive zero-order term. Since $z > 0$ in U and $z = 0$ on ∂U ($u = v = 0$ on ∂M ; $z = 0$ on the interior part of ∂U by continuity), the weak maximum principle [7, Cor. 3.2], again in the form of Appendix B(e), gives $\sup_U z \leq \sup_{\partial U} z^+ = 0$, a contradiction. Thus $u \geq v$.

Similarly let $U' := \{x \in M^\circ : u(x) > M_*\}$; on ∂M , $u = 0 < M_*$, so $\overline{U'} \cap \partial M = \emptyset$. On U' , $\theta(x, u) = M_*$, so $z := u - M_*$ satisfies

$$-\Delta z + Kz = a|\nabla u| + g(x, M_*) + KM_* - KM_* = a|\nabla z| + g(x, M_*) \leq a|\nabla z|,$$

because $g(x, M_*) = M_*(b(x) - c(x)M_*^{p-1}) \leq 0$. The same maximum principle on U' , where $z > 0$ and $z = 0$ on $\partial U'$, forces $U' = \emptyset$. Thus $u \leq M_*$.

Step 4: conclusion. Since $v \leq u \leq M_*$, $\theta(x, u(x)) = u(x)$ and (5) reduces to $-\Delta u + Ku = a|\nabla u| + g(x, u) + Ku$, i.e. u solves (2). The Schauder step of Theorem 3.12 gives $u \in C^{2,\alpha}(\overline{M})$, and $u \geq \varepsilon \phi_1 > 0$ in M° . Set $\Phi := u$. $\square$

What changed, and the relation to the sub-/super-solution literature. The earlier version of this proof used the super-solution $K\phi_1^2$. That construction fails for two independent reasons. First, $-\Delta(\phi_1^2) = 2\phi_1(-\Delta\phi_1) - 2|\nabla\phi_1|^2$: the gradient term enters with a negative sign, which reverses the boundary behaviour claimed there (for $\phi_1 = \sin x$ on $(0, \pi)$, $-\Delta(\phi_1^2) = -2\cos 2x < 0$

near the boundary). Second, no constant K makes $K\phi_1^2 \geq \varepsilon\phi_1$ hold up to a Dirichlet boundary: the inequality requires $K\phi_1 \geq \varepsilon$, which fails wherever $0 < \phi_1 < \varepsilon/K$. A constant super-solution has neither problem. The barrier pair $(\varepsilon\phi_1, M_*)$ is the classical one for problems with gradient dependence of at most quadratic growth, going back to Amann and Crandall [14] and the survey [15]; Steps 2–3 above give a self-contained proof in the present setting rather than an appeal to a general theorem, so that every hypothesis used is visible.

3.4 Master equation and the driven/coupled case

Remark 3.17 (Master equation and specializations). For generality one may consider the master form

$$-\Delta\Phi = a(x)|\nabla\Phi| + b(x)\Phi - c(x)\Phi^p + f(x), \quad \Phi|_{\partial M} = 0,$$

where f encodes exogenous significance / environmental input.

- **Autonomous (closed) case:** $f \equiv 0$, yielding (2).
- **Driven (coupled) case:** $b \equiv 0$, $c \equiv 0$, yielding a forced, interaction-sustained presence model.

Remark 3.18 (Forcing as the cost of striving). The exogenous forcing function $f(x)$ admits a novel interpretation connecting the master equation to the weak/strong coherence distinction. Suppose Φ_{weak} is a weak coherent configuration (a solution of the autonomous PDE; its dynamical stability is not established here, since no evolution law has been specified) and Φ_{strong} is the idealized strong coherence target. Then $f(x)$ may be modeled as the *measurable residual*—the energy required to pull the system from its stable weak configuration toward the strong coherence ideal.

If the gap between strong and weak coherence is large, the required $f(x)$ is large; if they nearly coincide, $f(x)$ is negligible. This reframing provides a roadmap for empirical testing: measure the forcing required to achieve near-ideal coherence, and this quantifies the “cost of striving.” The master equation thus bridges the conceptual distinction (Section 4.1) with a measurable dynamical quantity. The word “energy” is interpretive here: f is a forcing term, not a quantity derived from a variational functional, and no such structure is asserted for (2). Proposition 4.11 shows that on a compact manifold with zero boundary data the strong ideal is not attained by any nonnegative C^1 field, so the gap it measures is never zero in that setting.

3.5 Exactness of the threshold, the effective potential, and what remains open

Proposition 3.19 (Exact threshold without a gradient term). *Assume the coefficient hypotheses of Definition 3.1 with $a \equiv 0$. Then (2) has a solution that is positive in M° if and only if $\lambda_1(-\Delta - b; M) < 0$. More precisely, if $\Phi \geq 0$, $\Phi \not\equiv 0$, is a classical solution, then $\lambda_1 < 0$.*

Proof. Sufficiency is Theorem 3.16. For necessity let $\Phi \in C^2(\overline{M})$, $\Phi \geq 0$, $\Phi \not\equiv 0$, solve $-\Delta\Phi = b\Phi - c\Phi^p$, $\Phi|_{\partial M} = 0$, and let $\phi_1 > 0$ in M° be the principal eigenfunction. Green’s second identity (both functions vanish on ∂M) gives

$$\int_M (b\Phi - c\Phi^p)\phi_1 = \int_M (-\Delta\Phi)\phi_1 = \int_M \Phi(-\Delta\phi_1) = \int_M \Phi(b + \lambda_1)\phi_1,$$

hence

$$\lambda_1 \int_M \Phi\phi_1 = - \int_M c\Phi^p\phi_1 < 0,$$

because $c \geq c_0 > 0$, $\phi_1 > 0$ in M° and $\Phi^p \not\equiv 0$. Since $\int_M \Phi\phi_1 > 0$, $\lambda_1 < 0$. □

Remark 3.20 (The gradient term breaks the converse). With $a \neq 0$ the identity becomes $\lambda_1 \int \Phi \phi_1 = \int a |\nabla \Phi| \phi_1 - \int c \Phi^p \phi_1$, whose sign is not determined, so the argument does not extend. With $a > 0$ the converse is false in the continuum as well: Proposition 3.21 exhibits a positive solution on an interval with $\lambda_1 = +\frac{1}{4}$, and Proposition 3.32 does the same for the finite-graph model of Section 3.6. The spectral condition is therefore exact in the regime $a \equiv 0$ and a sufficient condition in general; the linear indicator λ_1 must not be described as an exact nonlinear threshold outside that regime.

Proposition 3.21 (A positive continuum solution below the linear threshold). *On $M = [0, \pi]$ with $a \equiv 1$, $b \equiv \frac{3}{4}$, $c \equiv 1$ and $p = 2$ (canonical closure with $\kappa = \gamma = \mu = 1$ and $\lambda = \frac{1}{4}$), the problem*

$$-u'' = |u'| + \frac{3}{4}u - u^2 \quad \text{in } (0, \pi), \quad u(0) = u(\pi) = 0,$$

has a solution $u \in C^{2,\alpha}([0, \pi])$ with $u > 0$ in $(0, \pi)$, although $\lambda_1(-\frac{d^2}{dx^2} - \frac{3}{4}; (0, \pi)) = 1 - \frac{3}{4} = \frac{1}{4} > 0$.

Proof. An explicit subsolution. Let $t(x) := \min\{x, \pi - x\}$ and

$$\phi(x) := \frac{e^{(\pi/2-t(x))/2} \sin(t(x)/2)}{\sin(\pi/4)}.$$

On $[0, \pi/2]$, where $t = x$, direct differentiation gives

$$\phi'(x) = \frac{e^{(\pi/2-x)/2}}{2 \sin(\pi/4)} \left(\cos \frac{x}{2} - \sin \frac{x}{2} \right) \geq 0, \quad \phi''(x) = -\frac{e^{(\pi/2-x)/2}}{2 \sin(\pi/4)} \cos \frac{x}{2},$$

so that $-\phi'' = |\phi'| + \frac{1}{2}\phi$ there. At $x = \pi/2$: $\phi = 1$, $\phi' = 0$, $\phi'' = -\frac{1}{2}$. The even reflection about $\pi/2$ therefore matches value, first and second derivative, and the identity $-\phi'' = |\phi'| + \frac{1}{2}\phi$ is invariant under reflection; hence $\phi \in C^2([0, \pi])$ with ϕ'' Lipschitz, $\phi > 0$ in $(0, \pi)$, $\phi(0) = \phi(\pi) = 0$, $\max \phi = 1$, and $-\phi'' = |\phi'| + \frac{1}{2}\phi$ on $[0, \pi]$. Put $v := \phi/4$, so $0 \leq v \leq \frac{1}{4}$. Then

$$-v'' - (|v'| + \frac{3}{4}v - v^2) = \frac{1}{4}(|\phi'| + \frac{1}{2}\phi) - \frac{1}{4}|\phi'| - \frac{3}{16}\phi + \frac{1}{16}\phi^2 = v(v - \frac{1}{4}) \leq 0,$$

so v is a subsolution, C^2 , nonnegative, vanishing at the endpoints and positive inside.

A constant supersolution. $w := \frac{3}{4}$ satisfies $g(x, w) = \frac{3}{4}w - w^2 = 0$, i.e. $-w'' = 0 \geq |w'| + g(x, w)$, and $w \geq \max v$.

Existence. The pair (v, w) has every property listed in Step 1 of the proof of Theorem 3.16 (What Steps 2–4 use); those steps apply verbatim with $K := 1 + \max\{0, 2w - \frac{3}{4}\} = \frac{7}{4}$ and yield a classical solution u with $v \leq u \leq \frac{3}{4}$, hence $u > 0$ in $(0, \pi)$. The eigenvalue is $\pi^2/\pi^2 - \frac{3}{4} = \frac{1}{4}$. $\square$

Remark 3.22 (Status and interpretation). Proposition 3.21 is a classical construction; it is not part of the Lean development and no novelty in the literature on gradient-dependent semilinear problems is claimed. Its content for the framework is conceptual: the creative drive $a|\nabla \Phi|$ can sustain a positive equilibrium in a regime where the linear viability operator alone does not cross its threshold, so the indicator λ_1 is a sufficient criterion for self-maintained presence, and an exact one only without drive. The numerical companion reproduces the example: the finite-difference solver started from v converges to a positive branch with maximum ≈ 0.2727 between the barriers, in agreement with an independent collocation solve, while the discrete eigenvalue is ≈ 0.25 .

Remark 3.23 (Effective potential and the gain β). The numerical companion writes the reaction coefficient as $q(x) := \beta b_{\text{base}}(x)$ with a dimensionless gain $\beta > 0$, because $\kappa, \gamma, \mu \in [0, 1]$ while Dirichlet eigenvalues scale like L^{-2} . All statements in this section apply verbatim with $b := q$: the threshold is $\lambda_1(-\Delta - q; M) < 0$, the bound of Lemma 3.10 uses $\max q_+$, and $\beta = 1$ recovers the

base equation. What is *not* true is that negativity of $\lambda_1(-\Delta - \beta b; M)$ for some auxiliary β says anything about the equation with coefficient b : on $(0, \pi)$ with $a = 0$, $c = 1$, $p = 2$, $b = \frac{1}{2}$ and $\beta = 3$, one has $\lambda_1(-\Delta - \beta b) = -\frac{1}{2} < 0$ while $\lambda_1(-\Delta - b) = \frac{1}{2} > 0$, so by Proposition 3.19 the unscaled equation has no positive solution. The formal interface discussed in Appendix A quantifies over such auxiliary β ; the correct instantiation is $b := q$ with $\beta = 1$.

Remark 3.24 (Uniqueness and stability). The scaling argument formalized in Lean (`scaling_uniqueness`) shows that a solution positive at a point where $c > 0$ has no solution multiple $k\Phi$, $k > 1$; it says nothing about uniqueness among all positive solutions, which remains open (Section 6). Likewise, an equilibrium solver classifies solutions of the stationary problem; it does not establish dynamical stability or a bifurcation type, for which no evolution law has been specified here.

3.6 The finite-graph model (machine-checked)

The continuum theorems above rest on classical elliptic theory. The Lean 4 development (Appendix A) contains a complete, unconditional proof of the analogous positive-existence theorem for a finite-graph version of (2), in which every operator is defined from data. We state that model and theorem here, together with an explicit example, a maximum bound proved for it, and a counterexample to the converse. The finite-graph formulation and the proof route through finite-dimensional inverse positivity, positive principal eigendata and the order-theoretic fixed-point core were proposed by Andrew Edmark; the particular discretization, the Jacobi splitting and the edge condition below are choices of the formalization.

Definition 3.25 (Finite-graph CD problem). Let V be a finite set of vertices, $w : V \times V \rightarrow [0, \infty)$ symmetric weights (self weights allowed), $B \subset V$ a boundary set with interior $I := V \setminus B$, and coefficients $\kappa, \gamma, \mu : V \rightarrow [0, 1]$, $a := \kappa\gamma\mu$, $b : V \rightarrow \mathbb{R}$, $c : V \rightarrow (0, \infty)$, $p > 1$. Define

$$(L_G u)(x) := \sum_{y \in V} w(x, y)(u(x) - u(y)), \quad |\nabla_G u|(x) := \left(\sum_{y \in V} w(x, y)(u(y) - u(x))^2 \right)^{1/2}.$$

A function $u : V \rightarrow \mathbb{R}$ solves the graph problem if

$$L_G u(x) = a(x)|\nabla_G u|(x) + b(x)u(x) - c(x) \max\{u(x), 0\}^p \quad (x \in I), \quad u(x) = 0 \quad (x \in B).$$

Both sums run over all of V , so edges to B contribute through the zero boundary values; there is no vertex measure and no degree normalization. The *energy* is $E(u) := \frac{1}{2} \sum_{x, y} w(x, y)(u(x) - u(y))^2 - \sum_x b(x)u(x)^2$, and the *principal eigenvalue* λ_1^G is the minimum of E over functions vanishing on B with $\sum_x u(x)^2 = 1$; equivalently, the smallest eigenvalue of the interior block of $\text{diag}(d) - w - \text{diag}(b)$, where $d(x) = \sum_y w(x, y)$ is the full degree, boundary edges included. The *interior graph* joins distinct interior vertices of positive weight.

Theorem 3.26 (Positive solution on a finite graph; Lean: `SemioticGraph.exists_pos_graph`). Suppose the interior graph is connected, $\lambda_1^G < 0$, and the edge condition

$$a(x) \leq \sqrt{w(x, y)} \quad \text{and} \quad a(y) \leq \sqrt{w(x, y)} \quad \text{for all distinct } x, y \in I \text{ with } w(x, y) > 0$$

holds. Then the graph problem has a solution u with $u(x) > 0$ for every $x \in I$.

Corollary 3.27 (Unweighted graphs; Lean: `exists_pos_graph_of_unweighted`). If every weight is 0 or 1, the edge condition follows from $0 \leq a \leq 1$, so connectivity of the interior graph and $\lambda_1^G < 0$ suffice.

Proof route (formalized in CdFormal/Graph). (i) *Principal eigendata.* The unit sphere of functions vanishing on B is compact and, since $I \neq \emptyset$, nonempty, so E attains its minimum λ_1^G at some φ ; the first-order condition gives $L_G \varphi = b\varphi + \lambda_1^G \varphi$ on I . Because $E(|\varphi|) \leq E(\varphi)$, $|\varphi|$ is also a minimizer, and a zero of a nonnegative eigenvector at an interior vertex propagates to its neighbours; connectivity of the interior graph then forces $|\varphi| > 0$ on I . (ii) *Fixed-point map.* With $K > 0$, the equation at $x \in I$ is equivalent to $u(x) = F_K(u)(x)$, where

$$F_K(u)(x) := \frac{\sum_y w(x,y)u(y) + a(x)|\nabla_G u|(x) + (b(x) + K)u(x) - c(x) \max\{u(x), 0\}^p}{d(x) + K},$$

and $F_K(u) := 0$ on B ; the identity $(d(x) + K)(u(x) - F_K(u)(x)) = L_G u(x) - (a|\nabla_G u| + bu - c u_+^p)(x)$ makes the residual of the equation visible. On functions with values in $[0, M]$ that vanish on B , F_K is monotone once K dominates $a(x) \sum_y \sqrt{w(x,y)} + c(x)pM^{p-1} - b(x)$: raising u by δ at a neighbour y of x lowers $a(x)|\nabla_G u|(x)$ by at most $a(x)\sqrt{w(x,y)}\delta \leq w(x,y)\delta$, which the term $w(x,y)u(y)$ recovers—this is where the edge condition is used, and the only place. (iii) *Barriers.* $\varepsilon\varphi$ is a subsolution once $c(x)\varepsilon^{p-1} \leq -\lambda_1^G$ for all x ; the plateau equal to M on I and 0 on B is a supersolution once $a(x)^2/4 + b(x) \leq c(x)M^{p-1}$ for all x . The plateau is not constant: next to the boundary its Laplacian is $S = M \sum_{y \in B} w(x,y)$ and its gradient norm is $\sqrt{MS}$, and the completed square $a\sqrt{MS} \leq S + \frac{a^2}{4}M$ absorbs the gradient term. (iv) *Fixed point.* F_K maps the order interval between $\varepsilon\varphi$ and the plateau, a complete lattice, into itself, so the Knaster–Tarski consequence `monotone_fixed_point_between` yields a fixed point there; it solves the problem and is at least $\varepsilon\varphi > 0$ on I . All constants are constructed from the data. $\square$

Remark 3.28 (Existence versus iteration). In finite dimensions the monotone iterates $F_K^n(\varepsilon\varphi)$ are nondecreasing and bounded by the plateau, hence convergent, and by continuity of F_K their limit is a fixed point. This classical remark justifies the iteration used in the numerical companion, but no convergence rate is claimed, and the Lean development proves existence only. A run of the iteration is validated by its residual, never by the theorem.

Example 3.29 (The triangle; Lean: `SemioticGraph.triangle`). Let $V = \{0, 1, 2\}$ with $w \equiv 1$ (including self weights), $B = \{0\}$, $\kappa = \gamma = \mu = 1$ (so $a = 1$), $b = 2$, $c = 1$, $p = 2$. The interior graph is the single edge $\{1, 2\}$, the weights lie in $\{0, 1\}$, and $E(0, 1, 1) = -2 < 0$, so $\lambda_1^G < 0$ and Corollary 3.27 applies (`exists_pos_triangle`). Explicitly, the interior block of $\text{diag}(d) - w - \text{diag}(b)$ is $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$, so $\lambda_1^G = -1$. The function $u = (0, 2, 2)$ solves the problem: at each interior vertex $L_G u = 2$ and $|\nabla_G u| = 2$, and $2 = 1 \cdot 2 + 2 \cdot 2 - 1 \cdot 2^2$ (`triangle_isSolution`, `triangle_gradNorm`); the gradient term is active. Here $b = 2$ is a free admissible coefficient, not the canonical closure $\kappa\gamma - \lambda\mu$ with $\lambda > 0$. The constants of the proof for this example are $\varepsilon = \frac{1}{4}$, $M = 7.75$ and $K = 62.5$; the numerical companion reproduces $(0, 2, 2)$ from both barriers.

Proposition 3.30 (Maximum bound on graphs). *Let $u \geq 0$ solve the graph problem and let $M := \max_V u > 0$. Then*

$$M \leq \max_{x \in I} \left(\frac{(b(x) + a(x)^2/4)_+}{c(x)} \right)^{1/(p-1)}.$$

Proof. The maximum is attained at some $x \in I$ since $u = 0$ on B . Put $S := L_G u(x) = \sum_y w(x,y)(M - u(y)) \geq 0$. Since $0 \leq M - u(y) \leq M$,

$$|\nabla_G u|(x)^2 = \sum_y w(x,y)(M - u(y))^2 \leq M \sum_y w(x,y)(M - u(y)) = MS.$$

The equation at x and the inequality $a\sqrt{MS} \leq S + \frac{a^2}{4}M$ (the arithmetic–geometric mean inequality; Lean: `mul_sqrt_mul_le`) give

$$S = a(x)|\nabla_G u|(x) + b(x)M - c(x)M^p \leq S + \frac{a(x)^2}{4}M + b(x)M - c(x)M^p,$$

hence $c(x)M^{p-1} \leq b(x) + a(x)^2/4$, which implies the claim. $\square$

Remark 3.31 (The graph bound differs from the continuum bound). The continuum bound $(B/c_0)^{1/(p-1)}$ of Lemma 3.10 uses $\nabla u = 0$ at an interior maximum, which has no graph analogue. The graph bound is attained: with $V = \{0, 1\}$, $B = \{0\}$, $w(0, 1) = \frac{1}{4}$, $a = b = c = 1$, $p = 2$, the function $u = (0, \frac{5}{4})$ solves the graph problem ($L_G u(1) = \frac{5}{16}$, $|\nabla_G u|(1) = \frac{5}{8}$, and $\frac{5}{16} = \frac{5}{8} + \frac{5}{4} - \frac{25}{16}$), exceeds the continuum value $b/c = 1$, and equals the graph bound $1 + \frac{1}{4}$. Proposition 3.30 is proved here classically; it is not a declaration of the Lean development at the pinned revision, although its key inequality is.

Proposition 3.32 (Negativity of λ_1^G is not necessary). *Take the triangle of Example 3.29 but with $b = \frac{1}{2}$. Then $u = (0, \frac{1}{2}, \frac{1}{2})$ solves the graph problem and is positive at both interior vertices, while $\lambda_1^G = +\frac{1}{2}$.*

Proof. At each interior vertex $L_G u = \frac{1}{2}$, $|\nabla_G u| = \frac{1}{2}$, and $\frac{1}{2} = 1 \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} - (\frac{1}{2})^2$. The interior block of $\text{diag}(d) - w - \text{diag}(b)$ is $\begin{pmatrix} 3/2 & -1 \\ -1 & 3/2 \end{pmatrix}$, with smallest eigenvalue $\frac{1}{2}$. $\square$

This does not contradict Theorem 3.26, whose spectral hypothesis is sufficient; it shows that the converse fails once the gradient term is active, in line with Remark 3.20.

Remark 3.33 (Three models that must not be identified). The paper now involves three distinct objects: the continuum problem (2); the finite-graph problem of Definition 3.25, whose theorem is machine-checked; and the centered finite-difference discretization used by the numerical companion, whose gradient in one dimension is $|(u_{i+1} - u_{i-1})/(2h)|$. The last two differ: on a unit-spaced path with values $(0, 1, 0)$ the centered derivative at the middle vertex is 0 while $|\nabla_G u| = \sqrt{2}$; and on a nearest-neighbour grid with weights h^{-2} , $|\nabla_G u|^2 = h^{-2}((u_{i+1} - u_i)^2 + (u_{i-1} - u_i)^2)$ has leading term $2(u')^2$ at smooth nonstationary points, not $(u')^2$. This local expansion is not a convergence theorem, and rescaling the coefficients or normalizing the weights would change the model that was verified. Consequently the finite-difference computations are numerical evidence for the continuum problem, the graph computations reproduce the verified finite model, and neither is a verification of the other.

3.7 Temporal resilience and coherence debt: unpacking contradiction cost

The canonical viability closure

$$b(x) = \kappa(x)\gamma(x) - \lambda\mu(x) \tag{6}$$

introduces a parameter $\lambda > 0$ weighting contradiction as a reduction in viability. In the present section we clarify the two distinct “lambdas” introduced in Remark 3.5 and then propose a temporal refinement of the closure in which contradiction cost is no longer an undetermined constant but an *effective* quantity determined by temporal coherence dynamics.

Two lambdas. We distinguish:

1. $\lambda_1(\mathcal{L})$, the *principal Dirichlet eigenvalue* of a linear operator $\mathcal{L}$ (e.g. $\mathcal{L} = -\Delta - b$). In the saturated CD model, the spectral sign condition $\lambda_1(-\Delta - b; M) < 0$ is a viability indicator: more negative values correspond to stronger distance from the threshold $\lambda_1 = 0$.
2. λ in (6), a *closure parameter* multiplying contradiction μ . In (6), increasing λ decreases b for fixed $\mu > 0$ and thus makes spectral viability harder to satisfy. Consequently, λ functions as a *contradiction cost/sensitivity parameter* (high λ = contradiction costly/fragile; low λ = contradiction less costly/resilient).

Figure 2 summarises the closure developed below and its feedback into the operator of Figure 1.

Temporal witness function and resilience factor

Let $\psi(t) \in [0, 1]$ denote a *temporal witness function* measuring instantaneous coherence at time t , with higher values indicating greater coherence and temporal integration. We assume $\psi \in C^3$ on the time interval considered, or, for sampled data, that ψ' , ψ'' and ψ''' denote declared finite-difference (or smoothing-spline) estimators at a stated sampling step; the quantities below are not defined otherwise. Following a temporal-coherence framework developed in companion work, we interpret derivatives as dynamical signatures: $\psi'(t)$ (volatility), $\psi''(t)$ (stabilizing vs destabilizing acceleration), and $\psi'''(t)$ (jerk/rupture).

Definition 3.34 (Base temporal resilience factor). Fix constants $\alpha_1, \alpha_3 > 0$ and $\beta \geq 0$; since ψ is dimensionless, α_1 , β and α_3 carry the units of time, time² and time³ respectively (or time is nondimensionalized by a stated reference scale). Define the *base temporal resilience factor*

$$R_{\text{base}}(t) := \frac{1 + \beta[\psi''(t)]^+}{1 + \alpha_1|\psi'(t)| + \alpha_3|\psi'''(t)|}, \quad [u]^+ := \max\{u, 0\}. \quad (7)$$

Remark 3.35 (Interpretation). The numerator rewards “acceleration toward coherence” (as encoded by $\psi'' > 0$ in the $\psi \in [0, 1]$ convention), while the denominator penalizes volatility and jerk. Thus R_{base} is large when coherence dynamics are smooth and stabilizing, and small when coherence changes rapidly or discontinuously.

Spectral capacity witness and the capacity envelope

A key architectural principle is that the system has a *sustainable capacity* to maintain coherent presence given its current geometry/boundary conditions and coefficient fields. We encode this capacity spectrally.

Let $M(t)$ denote the effective semiotic manifold at time t , allowing for the possibility that the operative system boundary/blanket changes with environment (e.g. infrastructure, community, housing). In practice, this can be represented equivalently by changes in effective boundary conditions, domain, or coefficient fields; we do not assume these are separable contributions.

Definition 3.36 (Spectral capacity witness). Let $b(\cdot, t)$ denote the viability potential at time t and define the principal eigenvalue

$$\lambda_1(t) := \lambda_1(-\Delta - b(\cdot, t); M(t)),$$

with Dirichlet boundary conditions on $\partial M(t)$. Fix a scale parameter $s > 0$ and define the *spectral capacity witness*

$$\psi_s(t) := \frac{1}{1 + \exp(\lambda_1(t)/s)} \in (0, 1). \quad (8)$$

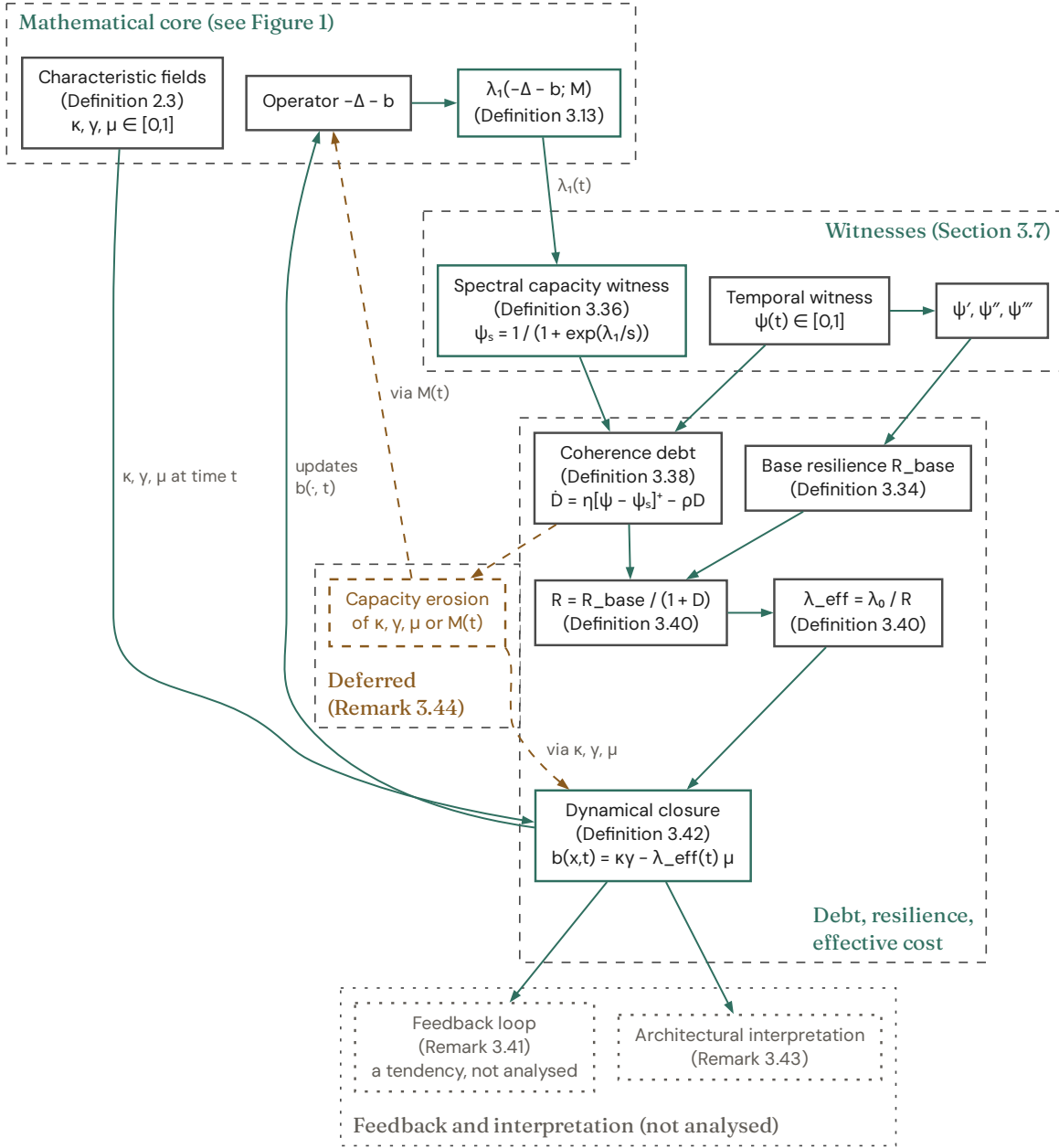


Figure 2: The temporal closure of Section 3.7. The derivatives of the empirical witness ψ enter the base resilience (Definition 3.34), and ψ itself enters the debt; the spectral capacity witness ψ_s (Definition 3.36) is computed from the principal eigenvalue of the current operator; only operation above capacity, $\psi > \psi_s$, feeds the coherence debt (Definition 3.38), which rises only while $D < (\eta/\rho)[\psi - \psi_s]^+$ and is bounded by $\max\{D(0), \eta/\rho\}$ (Remark 3.39); the debt lowers the total resilience and raises the effective contradiction cost (Definition 3.40); the effective cost and the fields κ, γ, μ at time t enter the dynamical closure (Definition 3.42). The closure supplies $b(\cdot, t)$, so the operator and $\lambda_1(t)$ are re-evaluated. The loop is a tendency, not an analysed dynamical system; the dashed capacity-erosion extension (Remark 3.44) is deferred.

Remark 3.37 (Meaning of ψ_s). The mapping (8) converts the real-valued spectral margin $\lambda_1(t)$ into a bounded witness comparable to $\psi(t)$: more negative λ_1 (deeper viability) yields ψ_s closer to 1, while λ_1 near or above 0 yields ψ_s near 1/2 or below. Other monotone maps $\mathbb{R} \rightarrow (0, 1)$ are possible; (8) is chosen for smoothness.

We interpret $\psi_s(t)$ as a *capacity envelope*: what the current system can sustain given its current structural conditions. This is a proposed proxy; the sigmoid score is not shown to be an upper envelope of an independently measured coherence witness ψ , and the comparison $\psi > \psi_s$ below is meaningful only under a stated calibration of the two scales. In contrast, $\psi(t)$ is an empirical witness of what the system is actually doing.

Coherence debt: chronic operation beyond capacity

The central burnout/recovery phenomenon motivating this section is the possibility that a system can temporarily *perform coherence beyond sustainable capacity*. Formally, this corresponds to $\psi(t) > \psi_s(t)$. Acute excursions are not necessarily pathological; the pathological regime is *sustained* overcapacity operation. We encode this temporally via a debt variable.

Definition 3.38 (Coherence debt dynamics). Fix parameters $\eta, \rho > 0$. Define the *coherence debt* $D(t) \geq 0$ as the solution of

$$\dot{D}(t) = \eta[\psi(t) - \psi_s(t)]^+ - \rho D(t), \quad D(0) \geq 0. \quad (9)$$

Remark 3.39 (Interpretation: asymmetric cost of mismatch). The positive-part term in (9) implements a structural asymmetry: only the regime $\psi > \psi_s$ (performing above capacity) accumulates debt. The decay term ρD models recovery/repayment when the system is no longer overextended. Equation (9) is linear in D and has the explicit solution

$$D(t) = e^{-\rho t} D(0) + \eta \int_0^t e^{-\rho(t-s)} [\psi(s) - \psi_s(s)]^+ ds,$$

an exponentially weighted integral of past overcapacity: it captures the chronicity of burnout (“debt that builds over time”) rather than a pointwise discrepancy. Because $\psi, \psi_s \in [0, 1]$, the excess is at most 1 and

$$0 \leq D(t) \leq e^{-\rho t} D(0) + \frac{\eta}{\rho} (1 - e^{-\rho t}) \leq \max \left\{ D(0), \frac{\eta}{\rho} \right\}.$$

Debt therefore *accumulates* only while the source $\eta[\psi - \psi_s]^+$ exceeds the repayment ρD ; during overcapacity operation it rises when $D < (\eta/\rho)[\psi - \psi_s]^+$ and falls otherwise, and it never exceeds η/ρ once below it. These are a priori identities for any solution with the stated witness ranges, including when ψ_s depends on D through the closure below; they are not a well-posedness theorem for the coupled system. (The integral formula is a direct consequence of the variation-of-constants formula for (9).)

From temporal resilience to effective contradiction cost

We now interpret the contradiction weight in (6) as an effective cost determined by temporal resilience and debt.

Definition 3.40 (Total resilience and effective contradiction cost). Let $\lambda_0 > 0$ denote baseline contradiction cost. Define

$$R(t) := \frac{R_{\text{base}}(t)}{1 + D(t)}, \quad (10)$$

and the *effective contradiction cost*

$$\lambda_{\text{eff}}(t) := \frac{\lambda_0}{R(t)}. \quad (11)$$

Remark 3.41 (Burnout spiral as feedback). When $\psi > \psi_s$ persists, debt $D(t)$ accumulates while $D < (\eta/\rho)[\psi - \psi_s]^+$ (Remark 3.39), reducing $R(t)$ and therefore increasing $\lambda_{\text{eff}}(t)$. Through the viability closure below, this reduces $b(\cdot, t)$, which pushes the spectral indicator $\lambda_1(t)$ upward (toward 0) and lowers $\psi_s(t)$. Thus sustained overcapacity operation is a self-reinforcing fragility *tendency*: capacity declines while debt accumulates. By the bound in Remark 3.39 the loop is neither automatically monotone nor unbounded; whether it converges, oscillates or saturates depends on the parameters and on the closure, which is not analysed here.

Dynamical viability closure

We replace the constant contradiction weight in (6) by $\lambda_{\text{eff}}(t)$.

Definition 3.42 (Dynamical viability closure). Define the viability potential as

$$b(x, t) := \kappa(x, t)\gamma(x, t) - \lambda_{\text{eff}}(t)\mu(x, t), \quad (12)$$

where λ_{eff} is given by (11).

Remark 3.43 (Architectural interpretation). The dynamical closure (12) makes explicit that contradiction cost is not a free constant but a compressed representation of temporal stability dynamics:

- smooth, stabilizing coherence trajectories (large R_{base}) reduce effective contradiction cost,
- chronic overcapacity operation (large D) increases effective contradiction cost,
- changes in the effective system boundary/blanket (encoded by $M(t)$ and the fields on it) shift the capacity envelope ψ_s through $\lambda_1(t)$.

This provides a principled bridge between temporal coherence (as measured by ψ) and viability in the CD PDE model.

Capacity erosion (deferred extension)

The model above treats debt D as a penalty on resilience but does not allow debt to permanently degrade the structural fields that generate capacity. In many real systems, prolonged overcapacity operation can erode care/coherence or increase contradiction exposure (“burning the furniture to heat the house”).

Remark 3.44 (Capacity erosion as a future-work extension). A natural extension is to let prolonged debt degrade the characteristic fields or effective system definition. For example, one may posit thresholded degradation laws such as

$$\dot{\kappa} = -\delta_{\kappa}[D - D_{\text{crit}}]^+, \quad \dot{\gamma} = -\delta_{\gamma}[D - D_{\text{crit}}]^+, \quad \dot{\mu} = +\delta_{\mu}[D - D_{\text{crit}}]^+,$$

with κ, γ, μ projected (clipped) onto $[0, 1]$, the range of Definition 2.3, or alternatively encode erosion by time-dependent boundary/geometry changes in $M(t)$. We do not analyze such extended dynamics here; the present contribution is the minimal debt-resilience closure (9)–(12).

4 Interpretive/structural layer

This section is conditional by design: it collects definitions and geometric statements that connect the PDE layer to the broader Creative Determinant framework.

4.1 Weak vs. strong coherence (logical status)

Definition 4.1 (Pointwise CD constraint (strong coherence)). A *strong coherent configuration* is a field Φ satisfying

$$\Phi(x) = |\nabla \Phi(x)| \kappa(x) \gamma(x) \mu(x) \quad \text{for all } x \in M.$$

Definition 4.2 (Weak coherence). A *weak coherent configuration* is a solution Φ of an associated PDE model (e.g. (2)) with stated boundary conditions.

Remark 4.3 (The generative gap). The relationship between weak and strong coherence is not merely an open mathematical problem—it is a *structural feature* of the framework. Strong coherence represents an idealization that real systems approximate but cannot fully achieve. Weak coherence captures the stable configurations that systems actually reach. The *distance* between them measures the cost of striving—the energy required to pull a system from its achievable state toward the ideal.

This perspective transforms an apparent limitation into a research program: Under what conditions is the gap small? When is it irreducible? How does the gap evolve under perturbation? The framework is defined by this necessary tension between the ideal and the messy achievable configurations—a commitment that aligns with enactivist insights about cognition as ongoing sense-making rather than static representation [2].

4.2 Conditional conservation of blessing–contradiction coupling

Definition 4.4 (Blessing–contradiction 1-form). Let $\Phi, \mu \in C^1(M)$. Define the 1-form

$$\omega := \Phi d\mu - \mu d\Phi.$$

Proposition 4.5 (Conditional conservation under equilibrium ansatz). *Let $U \subseteq M$ be open, and assume an equilibrium ansatz on U : there exists a C^1 function $g : [0, 1] \rightarrow \mathbb{R}$ such that*

$$\Phi(x) = g(\mu(x)) \quad \text{for all } x \in U.$$

Then ω is exact on U : with $G(m) := 2 \int_0^m g(s) ds - m g(m)$,

$$\omega = d(G(\mu)) \quad \text{on } U.$$

Consequently $\int_\Gamma \omega = 0$ for every closed piecewise- C^1 loop $\Gamma \subset U$, and in particular $\int_{\Gamma_1} \omega = \int_{\Gamma_2} \omega$ for any two loops in U (homotopic or not).

Proof. Under the ansatz, $d\Phi = g'(\mu) d\mu$ on U , so

$$\omega = \Phi d\mu - \mu d\Phi = (g(\mu) - \mu g'(\mu)) d\mu.$$

Since $G'(m) = 2g(m) - g(m) - m g'(m) = g(m) - m g'(m)$, we get $\omega = G'(\mu) d\mu = d(G(\mu))$, where $G \circ \mu \in C^1(U)$ because $g \in C^1$ and $\mu \in C^1$. The integral of an exact 1-form with a C^1 primitive over a closed loop vanishes by the fundamental theorem of calculus; no exterior derivative of ω (whose coefficients are only continuous) is needed. $\square$

Exactness versus closedness. Exactness is a strengthening of closedness: closedness alone gives equality of integrals over *homotopic* loops, exactness gives vanishing over every loop in U . The ansatz is not inferred from the PDE; it is a hypothesis about a phase, as stated in the next remark.

Remark 4.6 (Interpretation). The ansatz $\Phi = g(\mu)$ expresses a local functional dependence of presence on contradiction, i.e. a phase where the presence field is slaved to contradiction level sets. This is a strong idealization and is stated explicitly as such.

4.3 Care-weighted homotopy/energy principle (geometric interpretation)

Definition 4.7 (Care-weighted metric). On the open set $\{\kappa > 0\} \subseteq M$, define the conformally rescaled metric

$$g_\kappa := \kappa^{-1}g.$$

Definition 4.8 (Care-weighted homotopy energy). Let $H : S^1 \times [0, 1] \rightarrow M$ be a homotopy such that $H(S^1 \times [0, 1]) \subset \{\kappa > 0\}$. Define the energy

$$E_\kappa[H] := \int_0^1 \int_{S^1} \left\| \frac{\partial H}{\partial t} \right\|_{g_\kappa}^2 ds dt.$$

Proposition 4.9 (Care as a sufficient obstruction to resolution). *Let $\gamma_0 : S^1 \rightarrow M$ be a loop. If every homotopy contracting γ_0 to a point must intersect the zero-care set $\{\kappa = 0\}$, then there is no contracting homotopy H with finite $E_\kappa[H]$.*

Proof. The energy E_κ is defined only for homotopies with image in $\{\kappa > 0\}$ (Definition 4.8). If every contracting homotopy of γ_0 meets $\{\kappa = 0\}$, then no contracting homotopy is admissible, so in particular none has finite E_κ . $\square$

Remark 4.10 (Status: domain exclusion is not energy divergence). This is a geometric sufficient obstruction statement, not an “iff” characterization: we do not claim that finite-energy homotopy exists whenever a loop is null-homotopic in $\{\kappa > 0\}$, since that requires additional geometric hypotheses. It is also a statement about the admissible class, not a quantitative divergence theorem: it does not say that every extension or regularization of the energy blows up on homotopies that touch $\{\kappa = 0\}$. On the unit disk with $\kappa(x) = |x|$, the homotopy $H(s, t) = (1 - t)^2(\cos s, \sin s)$ contracts the boundary circle and meets $\{\kappa = 0\}$ only at $t = 1$; with the regularized weight $(\kappa + \epsilon)^{-1}$ its energy is

$$E_\epsilon = \int_0^1 \int_{S^1} \frac{4(1-t)^2}{(1-t)^2 + \epsilon} ds dt = 8\pi \left(1 - \sqrt{\epsilon} \arctan \frac{1}{\sqrt{\epsilon}} \right) \rightarrow 8\pi < \infty.$$

H is not admissible for E_κ , so this is not a counterexample to the proposition; it shows that whether care acts as an *energetic* barrier, rather than a topological one, depends on the rate at which κ vanishes and on the class of homotopies allowed, which the present definition fixes by fiat.

4.4 The strong ideal is unattainable by nonnegative fields

Proposition 4.11 (Obstruction to strong coherence). *Let M be compact, $\Phi \in C^1(\overline{M})$, and assume either $\partial M \neq \emptyset$ and $\Phi|_{\partial M} = 0$, or $\partial M = \emptyset$. Let $a := \kappa\gamma\mu$ be bounded.*

(a) *If $\Phi(x) = a(x)|\nabla\Phi(x)|$ for all $x \in M$, then $\Phi \equiv 0$. The zero field satisfies the constraint trivially; no nonzero field, of either sign, does.*

(b) For every such Φ , $\|\Phi - a|\nabla\Phi|\|_{L^\infty(M)} \geq \|\Phi\|_{L^\infty(M)}$.

Proof. (b) If $\Phi \equiv 0$ there is nothing to prove. Otherwise $|\Phi|$ attains its maximum at some x_0 with $\Phi(x_0) \neq 0$; x_0 is an interior point (in the first case because $\Phi = 0$ on ∂M , in the second because there is no boundary), and Φ has a local extremum there, so $\nabla\Phi(x_0) = 0$. Hence $(\Phi - a|\nabla\Phi|)(x_0) = \Phi(x_0)$, whose absolute value is $\|\Phi\|_\infty$. (a) follows: a strong coherent configuration has residual zero, so $\|\Phi\|_\infty = 0$. $\square$

Remark 4.12 (Interpretation). On a bounded semiotic manifold with dissipative boundary, or on a closed one, the strong ideal cannot be realized by any nonzero C^1 presence field: strong coherence is not merely unreached but unreachable in this setting except by the empty configuration, and only weak coherent configurations are attainable. Part (b) makes the gap quantitative in the right units: measured relative to the size of the field, the pointwise residual of the ideal is at least one, $\|\Phi - a|\nabla\Phi|\|_\infty / \|\Phi\|_\infty \geq 1$. The normalization matters, since scaling a field to $\epsilon\Phi$ makes the *absolute* residual tend to zero without bringing the field any closer to the ideal. This sharpens the generative gap of Remark 4.3: the “work of living” is not a finite distance to a reachable ideal but a tension that is irreducible relative to the field’s own scale. The proposition uses compactness, the C^1 regularity and the vanishing boundary data; it does not extend by itself to noncompact, nonsmooth or driven settings, where the ideal may remain a meaningful limit, and adding a forcing term to the separate equation (2) does not change this pointwise obstruction.

5 Operational layer

This section proposes testable conditions connecting the CD formalism to measurable dynamical systems.

5.1 The Creative Determinant condition

Definition 5.1 (Creative Determinant condition $CD(\varepsilon, \delta)$). Let $F : X \rightarrow X$ be a C^1 map on a smooth manifold X equipped with a fixed reference volume form (on $\mathbb{R}^n$, Lebesgue measure; on a Riemannian manifold, its volume form), and let $\varphi : X \rightarrow \mathbb{R}_{>0}$ be a pre-specified coherence observable. Fix a sampling population P : a finite ensemble of states, or a trajectory window $\{x, F(x), \dots, F^m(x)\}$, specified in advance. On the sample define

$$Y(x) := \log |\det \nabla F(x)|, \quad Z(x) := \log \varphi(F(x)),$$

where $\det \nabla F$ is the Jacobian determinant with respect to the reference volume form, and require the *validity assumptions*: $\nabla F(x)$ is invertible at every sampled x (points with $\det \nabla F = 0$, where $Y = -\infty$, must be excluded by a pre-registered rule), and neither Y nor Z is constant on P , so that both empirical variances are positive and the empirical correlation is defined. We say (F, φ, P) satisfies $CD(\varepsilon, \delta)$ if

$$\text{Corr}_P[Z, Y] \geq 1 - \varepsilon \quad \text{and} \quad \text{Var}_P[Y] > \delta.$$

The empirical regression slope $\hat{\alpha} := \text{Cov}_P[Z, Y] / \text{Var}_P[Y]$ is a separate, reportable quantity: it is the scale at which Z tracks Y , whereas the correlation is scale-free.

Why the scale is a separate quantity. An earlier formulation wrote $\text{Corr}[Z, \alpha Y]$ with a parameter α . Since $\text{Corr}[Z, \alpha Y] = \text{Corr}[Z, Y]$ for every $\alpha > 0$, $= -\text{Corr}[Z, Y]$ for $\alpha < 0$, and is undefined for $\alpha = 0$, the magnitude of α is not identifiable from a correlation condition, and no

inference about “ $\alpha \approx 1$ ” can rest on it. The slope $\hat{\alpha}$ above carries that information instead, and a claim about its value is a distinct empirical hypothesis to be preregistered separately.

Remark 5.2 (Connection to Lyapunov theory). By Oseledets’ multiplicative ergodic theorem [16], the time average of $Y = \log |\det \nabla F|$ along a trajectory converges (almost surely, for an ergodic invariant measure) to the sum of the Lyapunov exponents [17, 18]. Three quantities must be kept apart: the one-step statistic $Y(x)$ used in Definition 5.1; the *accumulated* volume change $\sum_{k < m} Y(F^k x) = \log |\det \nabla F^m(x)|$, whose average is what Oseledets’ theorem controls; and the Lyapunov spectrum itself, which requires a separate dynamical analysis. The CD condition is intended to detect regimes where a coherence observable tracks phase-space volume deformation step by step; a cumulative version replaces Y by the accumulated sum over a preregistered window. Fractal attractor structure and its Kaplan–Yorke dimension [19] are properties of the Lyapunov spectrum and are not inferred from a correlation between Z and Y ; establishing them is independent dynamical evidence, not a consequence of the CD condition.

5.2 Falsifiability protocols (empirical proposals)

We list falsifiability criteria as operational proposals; they are not derived from the PDE theorem.

Definition 5.3 (Framework falsification (empirical)). The CD framework is empirically falsified if any of the following are demonstrated:

- (F1) Coherence without coupling: stable high coherence with near-zero (or negative) correlation $\text{Corr}_P[Z, Y]$ between the coherence observable and the Jacobian volume statistic, under the validity assumptions of Definition 5.1.
- (F2) Conservation violation: statistically significant disagreement of $\int_\Gamma \omega$ between homotopic paths within a verified equilibrium region.
- (F3) Hamiltonian coherence: volume-preserving dynamics exhibiting the same coherence–exploration coupling as dissipative systems.
- (F4) Resolution without care: contradiction resolution along paths confined to low-/zero-care regions (as operationalized).
- (F5) Universal uniqueness: systematic absence of multiplicity where the heuristic non-uniqueness picture predicts multiple coherent configurations.

Remark 5.4 (Protocols). Concrete testing protocols for large language models and other high-dimensional systems can be specified by choosing: (i) a coherence observable φ (e.g. compressibility/entropy/intrinsic dimension proxies), (ii) a feasible estimator for $\log |\det \nabla F|$ (e.g. factorization across steps, trace estimators, or low-rank sketches), together with the reference coordinates in which the determinant is taken, (iii) the sampling population or time window and the rule for excluding singular Jacobians, and (iv) statistical thresholds for correlation and robustness checks. A negative result rejects the specified operational instantiation (F, φ, P) ; it does not bear on the elliptic theorems of Section 3, which are proved independently of any empirical interpretation.

6 Discussion and limitations

Mathematical contributions

The core mathematical results are:

1. **Existence framework (Theorem 3.12):** The autonomous viability model with saturation (2) has a compact fixed-point formulation with a priori bounds for nonnegative weak coherent configurations; the zero field is always a solution, so this theorem alone does not produce presence.
2. **Positive existence (Theorem 3.16):** When the viability threshold is exceeded ($\lambda_1(-\Delta - b; M) < 0$), there exists a solution positive in the interior, enclosed between $\varepsilon\phi_1$ and a constant. This gives a sufficient criterion for when coherent presence can be self-maintained, and by Proposition 3.19 an exact one when $a \equiv 0$.
3. **Finite-graph theorem (Theorem 3.26):** The analogous statement on finite weighted graphs, proved without analytic hypotheses and machine-checked, together with a maximum bound and a counterexample showing that the spectral condition is not necessary once the gradient term is active.

Mathematical limitations

The saturation term $-c\Phi^p$ is necessary to break positive homogeneity and enable nontrivial Dirichlet solutions. Without it (the V1 limiting case with $a \equiv 0$) a nonnegative nontrivial solution exists only when $\lambda_1(-\Delta - b; M) = 0$, and then only as a multiple of ϕ_1 ; generically only the trivial solution remains. This is a structural constraint, not a proof gap. Two questions are open: uniqueness of the positive solution (Remark 3.24) and the dynamical stability of equilibria, which requires an evolution law not specified here. The converse of the threshold with a gradient term is settled negatively in both the continuum (Proposition 3.21) and the finite-graph model (Proposition 3.32); what remains open there is a characterization of the positive-existence region for $a > 0$. The continuum theorems are classical proofs; their Lean counterparts are conditional on an analytic interface that the concrete operator does not instantiate (Appendix A).

Conceptual limitations

The pointwise CD constraint and the PDE proxy are not proven equivalent; this gap is treated as a structural feature (the “generative tension”) rather than a deficiency. The canonical viability closure $b = \kappa\gamma - \lambda\mu$ is a modeling choice grounded in observed support-versus-cost dynamics; alternative closures remain open for investigation.

Validating the linear closure

The canonical closure $b = \kappa\gamma - \lambda\mu$ is a modeling choice, but one that can be empirically tested. The accompanying computational notebook (`cd_pde_demo.ipynb`) sweeps the contradiction cost λ under the canonical closure and records where the positive branch gives way to the zero branch; it does not compare alternative closure forms, which remains future work.

Future work should:

1. Compare linear, multiplicative ($b = \kappa\gamma/(1 + \lambda\mu)$), and exponential ($b = \kappa\gamma e^{-\lambda\mu}$) closures.
2. Identify which closure best captures observed bifurcation points in real systems.
3. Use computation to validate the conceptual simplicity of the linear form.

The linear closure is not the final word—it is the *simplest hypothesis* against which alternatives must be tested.

Operational limitations

The CD condition and falsifiability protocols require careful pre-registration of observables and estimators to avoid tautological fitting; they are empirical proposals.

Relationship to existing frameworks

The CD framework shares conceptual DNA with several existing formalizations of self-organization:

- **Autopoiesis** [1]: CD’s “viability threshold” operationalizes the autopoietic notion of self-maintenance through boundary conditions.
- **Free Energy Principle** [20]: Both frameworks formalize how systems maintain coherence against environmental perturbation; CD emphasizes geometric/topological structure while FEP emphasizes information-theoretic bounds.
- **Enactivism** [21, 2]: The “semiotic manifold” reflects enactivism’s emphasis on meaning as emergent from organism-environment coupling.
- **Dissipative structures** [22]: The viability threshold condition is analogous to criticality conditions for far-from-equilibrium self-organization.

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A Formal Verification Details

A.1 Verification overview

The formalization [3] is a Lean 4 project against Mathlib, both at version v4.34.1, and this paper refers to its revision

ad65a646ec050acff52e03bc4bb81fd218eb2510.

Its continuous integration builds the root module and every tracked module with warnings treated as errors, runs the Mathlib environment linters, and executes `CdFormal/Verify.lean`, which prints the axioms of 49 selected declarations; each record may use only `propext`, `Classical.choice` and `Quot.sound`, and a separate check inspects the compiled environment to confirm that no module declares an `axiom`. The same steps were replicated locally at that revision for this version of the paper.

Three kinds of statement must be distinguished.

1. **Proved outright.** Algebraic, real-analytic and order-theoretic lemmas, and the complete finite-graph development of Section 3.6: operators, principal eigendata, barriers, fixed point, the positive-existence theorem, its unweighted corollary, and the explicit triangle example.
2. **Proved conditionally.** The two continuum existence theorems are derived from the hypotheses packaged in the class `PDEInfra` and the structures `SolutionOperator` and `PrincipaleEigendata`. These are structure fields, not Lean axioms, so `#print axioms` does not list them; they appear in the theorem signatures. They are assumptions about an abstract operator T on the bare function type $M \rightarrow \mathbb{R}$.

3. **Not formalized.** The classical analytic content of Section 3 (Schauder and L^q theory, maximum principles, the eigenfunction of $-\Delta - b$), the identification of $(\pi/L)^2 - \beta b$ with an eigenvalue, Lemma 3.11, Propositions 3.19, 3.21, 3.30 and 4.11, and everything in Section 3.7 and Sections 4–5.

A.2 The model and the interface

In Lean the manifold is a compact connected space with an analytic atlas modelled on $\mathbb{R}^n$ (so it has no manifold boundary), the Laplacian and gradient norm are fields of a structure `SemioticOperators` with a few algebraic properties (not built from a metric), and `SemioticBVP.boundary` is an arbitrary set with nonempty complement. The default equation is imposed at every point of M , boundary points included, and `SemioticBVP.equation` and `SemioticBVP.boundaryCondition` are overridable fields. Table 1 lists the five fields of `PDEInfra` and the classical result each stands in for.

Table 1: The five `PDEInfra` hypotheses (revision `ad65a64`).

Field	Lean statement (paraphrase)	Relation to this paper
<code>T_compact</code>	T maps von Neumann bounded subsets of $M \rightarrow \mathbb{R}$ (product topology) to sets with compact closure.	On $M \rightarrow \mathbb{R}$ both conditions reduce to pointwise boundedness; this is not the C^1 compactness of Lemma 3.7.
<code>linfty_bound</code>	For $B \geq b$ there is $K > 0$ with $ u \leq K$ for every $u = \tau T(u)$, $\tau \in [0, 1]$.	Corresponds to Lemma 3.10 for the truncated operator T ; false for the untruncated operator $\tilde{T}$ (Section 3.2, <i>What changed</i>).
<code>schaefer</code>	The two conditions above yield a fixed point of T .	Schaefer’s theorem also needs continuity on a Banach space, which is not assumed there.
<code>fixed_point_nonneg</code>	Fixed points of T are nonnegative.	Lemma 3.9 for the truncated T ; false for $\tilde{T}$.
<code>monotone_iteration</code>	For every β and supplied eigen-data with negative eigenvalue there is a fixed point of T positive at some interior point.	Quantifies over auxiliary β while the equation uses b ; overstrong for $\beta \neq 1$ (Remark 3.23). Theorem 3.16 is its $\beta = 1$ instance, with positivity everywhere in M° .

Collisions. The interface is not instantiated by the concrete operators of this paper, for three independent reasons recorded here so that the conditional theorems are not misread. (i) The field `T` of `SolutionOperator` acts on all of $M \rightarrow \mathbb{R}$, and `T_compact` uses the product topology; the concrete T of Section 3 is defined and compact only on a C^1 space. (ii) The default equation is asserted at boundary points too, whereas (2) imposes the equation in the interior with a separate boundary condition. (iii) The untruncated operator satisfies neither `linfty_bound` nor `fixed_point_nonneg` (Section 3.2, *What changed*). The Lean implications are correct conditional statements; the discharge of their hypotheses for the concrete problem is the classical analysis of Section 3, which is not formalized.

A.3 Paper-to-Lean alignment

Table 2 maps statements of this paper to declarations at the pinned revision (file paths relative to `CdFormal/`). “Proved” means the kernel checked the proof with no hypothesis beyond the stated

signature; “conditional” means proved from `PDEInfra` and `SolutionOperator`; “definition” means the declaration type-checks; “ingredient” means the declaration proves one step of the cited result.

Table 2: Paper-to-Lean alignment at revision `ad65a64`.

Paper	Lean declaration	File	Status
Def. 2.1 (semiotic manifold)	<code>SemioticManifold</code>	<code>Basic</code>	definition (abstract model)
Def. 2.3, Def. 3.1 (fields, $a = \kappa\gamma\mu$)	<code>SemioticContext</code> , <code>SemioticContext.a</code>	<code>Basic</code>	definition
Def. 3.3 (canonical closure)	<code>SemioticContext.canonicalViability</code>	<code>Basic</code>	definition
Def. 3.1 ($V1'$, weak coherent configuration)	<code>SemioticBVP</code> , <code>IsWeakCoherentConfiguration</code>	<code>Basic</code>	definition (overridable fields)
$a \in [0, 1]$, $p - 1 > 0$	<code>SemioticContext.a_nonneg</code> , <code>a_le_one</code> , <code>p_sub_one_pos</code>	<code>CoefficientLemmas</code>	proved
Operator algebra	<code>laplacian_zero</code> , <code>laplacian_linear</code> , <code>gradNorm_zero</code>	<code>OperatorLemmas</code>	proved
Zero solution (Section 3.2)	<code>zero_solves_equation</code>	<code>OperatorLemmas</code>	proved
Lem. 3.7	<code>PDEInfra.T_compact</code>	<code>Axioms</code>	hypothesis (different topology)
Lem. 3.9	<code>PDEInfra.fixed_point_nonneg</code>	<code>Axioms</code>	hypothesis
Lem. 3.10	<code>PDEInfra.linfty_bound</code> ; <code>linfty_bound_algebraic</code> , <code>rpow_le_of_mul_rpow_le</code>	<code>Axioms</code> ; <code>LinftyAlgebraic</code>	hypothesis; algebraic step proved
Lem. 3.11	—	—	not formalized
Thm. 3.12	<code>SemioticBVP.exists_isWeakCoherentConfiguration</code>	<code>Theorems</code>	conditional
Def. 3.13	<code>PrincipalEigendata</code>	<code>Axioms</code>	supplied data, not constructed
Thm. 3.16	<code>SemioticBVP.exists_pos_isWeakCoherentConfiguration</code>	<code>Theorems</code>	conditional; positive at one interior point
1D threshold, $b > 0$	<code>viabilityThreshold</code> , <code>spectral_characterization_id</code> , <code>viabilityThreshold_lt_iff</code>	<code>Theorems</code>	proved (arithmetic; eigenvalue identification classical)
Rem. 3.24 (proportional solutions)	<code>scaling_uniqueness</code> , <code>scaling_algebraic_contradiction</code>	<code>ScalingUniqueness</code> , <code>Theorems</code>	proved
Order-theoretic core	<code>monotone_fixed_point_between</code> , <code>OrderHom.nextFixed_le_of_le</code>	<code>MonotoneFixedPoint</code>	proved
Prop. 3.19, Prop. 3.21, Rem. 3.23	—	—	not formalized (classical)
Def. 3.25	<code>SemioticGraph</code> , <code>SemioticGraph.laplacian</code> , <code>gradNorm</code> , <code>IsSolution</code> , <code>energy</code> , <code>unitSphere</code> , <code>principalEigenvalue</code> , <code>interiorGraph</code>	<code>Graph/Basic</code>	definition
Thm. 3.26 (i)	<code>exists_pos_eigenvector</code> , <code>principalEigenvalue_neg</code> , <code>principalEigenvalue_mul_le</code> , <code>principalEigenvalue_le_of_laplacian_eq</code>	<code>Graph/Spectral</code>	proved
Thm. 3.26 (ii)	<code>fixedPointMap</code> , <code>mul_sub_number</code> , <code>fixedPointMap_eq_iff_isSolution</code> , <code>fixedPointMap_mono</code>	<code>Graph/FixedPoint</code>	proved
Thm. 3.26 (iii)	<code>smul_subsolution</code> , <code>plateau_supersolution</code> , <code>mul_sqrt_mul_le</code>	<code>Graph/FixedPoint</code>	proved
Thm. 3.26 (iv)	<code>exists_isSolution_between</code> , <code>exists_pos_graph</code>	<code>Graph/FixedPoint</code> , <code>Graph/Existence</code>	proved
Cor. 3.27	<code>exists_pos_graph_of_unweighted</code>	<code>Graph/Existence</code>	proved
Ex. 3.29	<code>triangle</code> , <code>exists_pos_triangle</code> , <code>triangle_isSolution</code> , <code>triangle_gradNorm</code>	<code>Graph/Example</code>	proved
Prop. 3.30	<code>mul_sqrt_mul_le</code>	<code>Graph/FixedPoint</code>	ingredient; the bound itself is not formalized
Prop. 3.32, Rem. 3.33	—	—	not formalized
Prop. 4.11; Section 3.7; Sections 4–5	—	—	not formalized

As Mathlib’s elliptic theory grows, the fields of `PDEInfra` could be replaced by proofs; doing so would also require constructing the operators from the metric and representing a manifold with

boundary. Until then, the continuum existence theorems are verified as implications, and their analytic content is the classical proof given in Section 3.

B Transfer of the cited Euclidean results to the manifold

The theorems cited from Gilbarg–Trudinger [7] in Section 3.2 are Euclidean statements. This appendix records how each is used on the compact, connected manifold M with smooth boundary of Definition 2.1; nothing here is new, but the passage from a bounded domain to M is not a single localization step, and we separate the parts that transfer by patching from the parts that do not.

Fix a finite atlas (U_i, x_i) whose charts map interior patches onto balls and boundary patches onto half-balls with ∂M flattened, and a subordinate partition of unity (χ_i) . In each chart $\Delta_g = g^{jk}\partial_{jk} + (\text{smooth})\partial_j$ is uniformly elliptic with coefficients in $C^\infty(\overline{x_i(U_i)})$, and $\Delta_g - K$ has zero-order coefficient $-K \leq 0$.

- (a) *Function spaces.* $W^{k,q}(M)$ and $C^{k,\beta}(\overline{M})$ are defined through the charts; different finite atlases give equivalent norms.
- (b) *Embeddings and interpolation* (Theorems 7.26 and 7.28 of [7]) hold on M by applying the Euclidean statements to $\chi_i u$ and summing over the finitely many charts; the Hölder quotient for points in different charts is bounded by $2\|u\|_\infty/\delta^\beta$ with δ a Lebesgue number of the cover. These steps are routine.
- (c) *A priori estimates.* Applying the interior and boundary $W^{2,q}$ estimates of [7, Ch. 9] to $\chi_i u$, using that $[\Delta_g, \chi_i]$ is a first-order operator, and absorbing $\|u\|_{W^{1,q}}$ by (b), gives $\|u\|_{W^{2,q}(M)} \leq C(\|Lu\|_{L^q(M)} + \|u\|_{L^q(M)})$ for $u \in W^{2,q}(M) \cap W_0^{1,q}(M)$, $L = \Delta_g - K$, $K \geq 0$; the same patching of the interior and boundary Schauder estimates of [7, Ch. 6] gives the $C^{2,\beta}$ bound. The $\|u\|_{L^q}$ term is removed exactly as in the proof of Lemma 9.17 of [7], by contradiction and compactness, once uniqueness is known: for $q > n$, $u \in W^{2,q} \cap W_0^{1,q} \subset C^1(\overline{M})$, and $Lu = 0$ gives $\int_M |\nabla u|^2 + Ku^2 = 0$ by Green's formula (valid for such u by approximation), hence $u = 0$ because M is connected and $u = 0$ on $\partial M \neq \emptyset$.
- (d) *Existence for the linear Dirichlet problem* is not an estimate and is obtained separately. Let $K \geq 0$. For $f \in L^2(M)$ the Lax–Milgram theorem gives a unique weak solution $u \in H_0^1(M)$ of $-\Delta_g u + Ku = f$ (the form is coercive because $\partial M \neq \emptyset$ gives a Poincaré inequality), and the difference-quotient method in charts (interior, and at a flattened boundary) gives $u \in H^2(M)$. If $f \in L^q(M)$ with $q \geq 2$, then $f \in L^2(M)$ and the L^q regularity theory of [7, Ch. 9] (Lemma 9.16, from which Theorem 9.15 is derived), applied to $\chi_i u$ on chart images chosen as $C^{1,1}$ domains, raises the exponent to q in finitely many Sobolev steps, because the commutator $[\Delta_g, \chi_i]u$ involves only u and ∇u ; hence $u \in W^{2,q}(M)$. If $1 < q < 2$, which can occur only when $n = 1$, then $L^q(M) \not\subset L^2(M)$ and the H^2 step is not available for f itself: approximate f in $L^q(M)$ by $f_k \in L^2(M)$, solve for $u_k \in H^2(M) \cap H_0^1(M) \subset W^{2,q}(M) \cap W_0^{1,q}(M)$ (M is compact and $q < 2$), and apply the estimate of (c) to the differences, $\|u_k - u_l\|_{W^{2,q}} \leq C\|f_k - f_l\|_{L^q}$; the sequence is Cauchy in $W^{2,q}(M)$, and its limit $u \in W^{2,q}(M) \cap W_0^{1,q}(M)$ solves $-\Delta_g u + Ku = f$. In both cases this is the manifold form of Theorem 9.15, and with (c) it yields $\|u\|_{W^{2,q}} \leq C\|f\|_{L^q}$, the manifold form of Lemma 9.17. For $f \in C^{0,\beta}(\overline{M})$ the same u lies in $C^{1,\gamma}(\overline{M})$ by (b), and the Hölder regularity of strong solutions with Hölder data, interior ([7, Ch. 9]) and at a flattened boundary ([7, Ch. 6]), applied in charts, gives $u \in C^{2,\beta}(\overline{M})$, the manifold form of Theorem 6.14; uniqueness in that class is again (c).

- (e) *Maximum principles.* The proofs of Theorem 3.1 and Corollary 3.2 of [7] use a global auxiliary function $e^{\gamma x_1}$ and do not localize. On an open set $U \subset M^\circ$ we use instead the strong maximum principle, Theorem 3.5 of [7], which is local (it is proved in balls) and requires only ellipticity, a bound on $|\beta|/\lambda$ and, when $c \leq 0$, on $|c|/\lambda$, with no continuity of the coefficients: if $w \in C^2(U) \cap C^0(\overline{U})$ satisfies $\Delta_g w + \beta \cdot \nabla w + cw \geq 0$ with $c \leq 0$ bounded, $w \leq 0$ on ∂U , and $\sup_U w > 0$, then the positive maximum is attained at an interior point x_0 ($\overline{U}$ is compact); Theorem 3.5 on the connected component V of U containing x_0 makes w constant on V , and since $\partial V \subset \partial U$ is nonempty (M is connected and $\partial M \neq \emptyset$) and $w \leq 0$ there, this is a contradiction. Hence $\sup_U w \leq \sup_{\partial U} w^+$, which is the form of Theorem 3.1 ($c = 0$) and Corollary 3.2 ($c \leq 0$) used in Lemma 3.9 and in Step 3 of the proof of Theorem 3.16. The point requiring care is the gradient coefficient $\beta = a\nabla z/|\nabla z|$, which is only bounded and measurable; this is admissible in Theorem 3.5, whose proof uses only the bounds.
- (f) *The principal eigenvalue* of $-\Delta_g - b$, $b \in C^{0,\alpha}(M)$. The Rayleigh-quotient argument is chart-free once the compact embedding $H_0^1(M) \hookrightarrow L^2(M)$ from (b) is available; the minimizer $\phi_1 \in H_0^1(M)$ is a weak solution of $-\Delta_g \phi_1 + K\phi_1 = (b + \lambda_1 + K)\phi_1$ with right-hand side in $L^2(M)$, so (d) with its exponent bootstrap gives $\phi_1 \in W^{2,q}(M) \subset C^{1,\gamma}(\overline{M})$, the right-hand side is then in $C^{0,\alpha}(\overline{M})$, and $\phi_1 \in C^{2,\alpha}(\overline{M})$; $|\phi_1|$ is also a minimizer, and the strong maximum principle (Theorem 3.5 of [7] on M° , in its minimum form: $c \leq 0$ and a nonpositive interior minimum force a constant) applied to $|\phi_1|$ with $c = -(b + \lambda_1)^-$ shows $\phi_1 > 0$ in M° ; two positive eigenfunctions cannot be L^2 -orthogonal, so λ_1 is simple. This is Theorem 8.38 of [7], whose hypotheses (self-adjoint divergence form, bounded measurable coefficients) are met, transferred to M .

Routine steps: (a), (b), the patching in (c), and the Rayleigh-quotient part of (f). Steps that need care: removing $\|u\|_{L^q}$ in (c), which needs uniqueness (the Green's-formula argument uses $q > n$ so that $u \in C^1(\overline{M})$); existence in (d), which glues three separate theories (Hilbert space, H^2 regularity, L^q bootstrap) and needs the approximation argument when $1 < q < 2$, since $L^q(M) \not\subset L^2(M)$ there; the weak maximum principle on an arbitrary open $U \subset M^\circ$ in (e), which must go through the strong one and argue on components; and the comparison with the gradient term in Step 3, which uses $c = -K \leq 0$, a sign condition that is essential.

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